Difference-in-Differences

Where We've Been & Where We're Going

Alex Cardazzi
Department of Economics

Talk Outline

- The Fundamental Problem of Causal Inference
- What is Difference-in-Differences ("DiD")?
- Two-Way Fixed Effects ("TWFE")
- Event Studies
- What's wrong with TWFE?
- Not covered: Triple Differences, Continuous Treatment, "Small" Treated Groups

Causal Inference

> Correlation does not imply causation.

Suppose we observe information on 50 units (states, indexed by s) over time (years, indexed by t) and their "treatment" status measured as $D \in \{0, 1\}$. In a perfect world, we want to know:

$$Y_{st}^{D=1} - Y_{st}^{D=0}$$

Then, we can take an average across all states and years:

$$\frac{1}{n} \sum Y_{st} = E[Y_{st}] \rightarrow E[Y_{st}^{D=1}] - E[Y_{st}^{D=0}] = ATE$$

However, we can only observe the outcome Y_{st} when $D = 1$ *or* $D = 0$, but not both. Instead, we need to rely on observational data. Then, we can calculate the average of Y_{st} when $D = 1$ and subtract the average of Y_{st} when $D = 0$.

$$E[Y_{st}|D_{st} = 1] - E[Y_{st}|D_{st} = 0]$$

Causal Inference

$$E[Y_{st}^{D=1}] - E[Y_{st}^{D=0}] \overset{?}{\approx} E[Y_{st}|D_{st} = 1] - E[Y_{st}|D_{st} = 0]$$

An example: I believe ice cream sales cause crime fluctuations. I can calculate the average crime rate (Y) when ice cream sales (D) are high and subtract the average crime rate when ice cream sales are low.

Of course, temperature is causing both crime and ice cream sales! Putting this into regression form does not make it any less invalid *until* I can control for temperature.

$$\text{Crime}_{st} = \beta_0 + \beta_1 \times \text{ICS}_{st} + \epsilon_{st}$$

As a note, *this* is why RCTs are the gold standard in research. Randomization in and of itself is not what is important, but rather, the fact that the randomization severs or breaks confounding correlations. If we could randomly assign ice cream sales, our regression could identify its effect on crime.

Causal Inference

So how do we identify causality when treatment is *endogenous*?

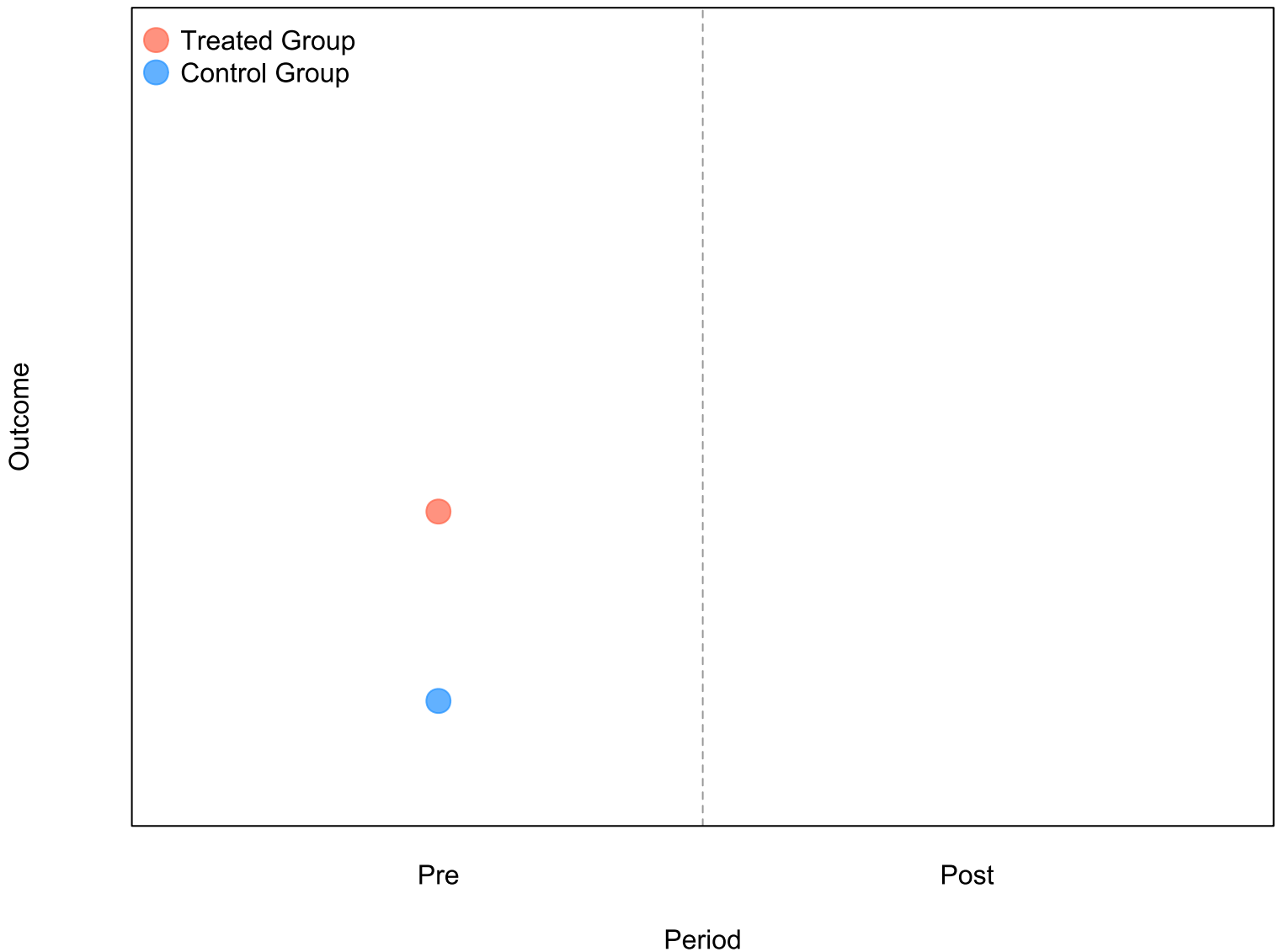
1. ~~We can control for all of the confounding factors.~~
2. ~~We can isolate exogenous variation in the treatment with an *instrumental variable*, thus breaking the correlations between the treatment and confounding.~~
3. We can explicitly estimate the counterfactual of Y_{st}.

What is Difference-in-Differences?

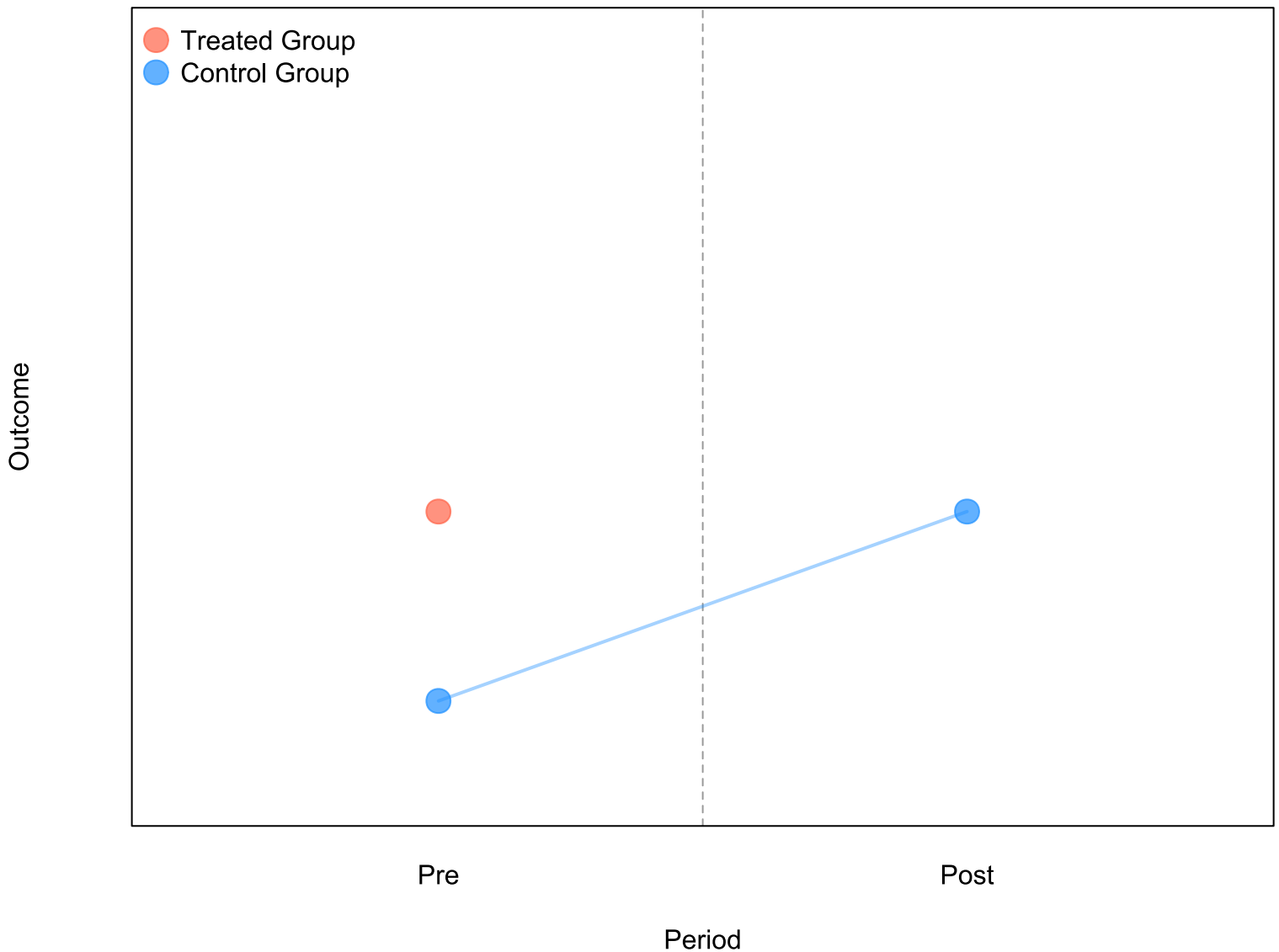
In the most simple terms, DiD compares changes in a treated group to changes in a control group after some treatment occurs.

The changes in the control group inform us of how the treated group would have changed *if it had not been treated*. In other words, we explicitly estimate a counterfactual using changes in the control group

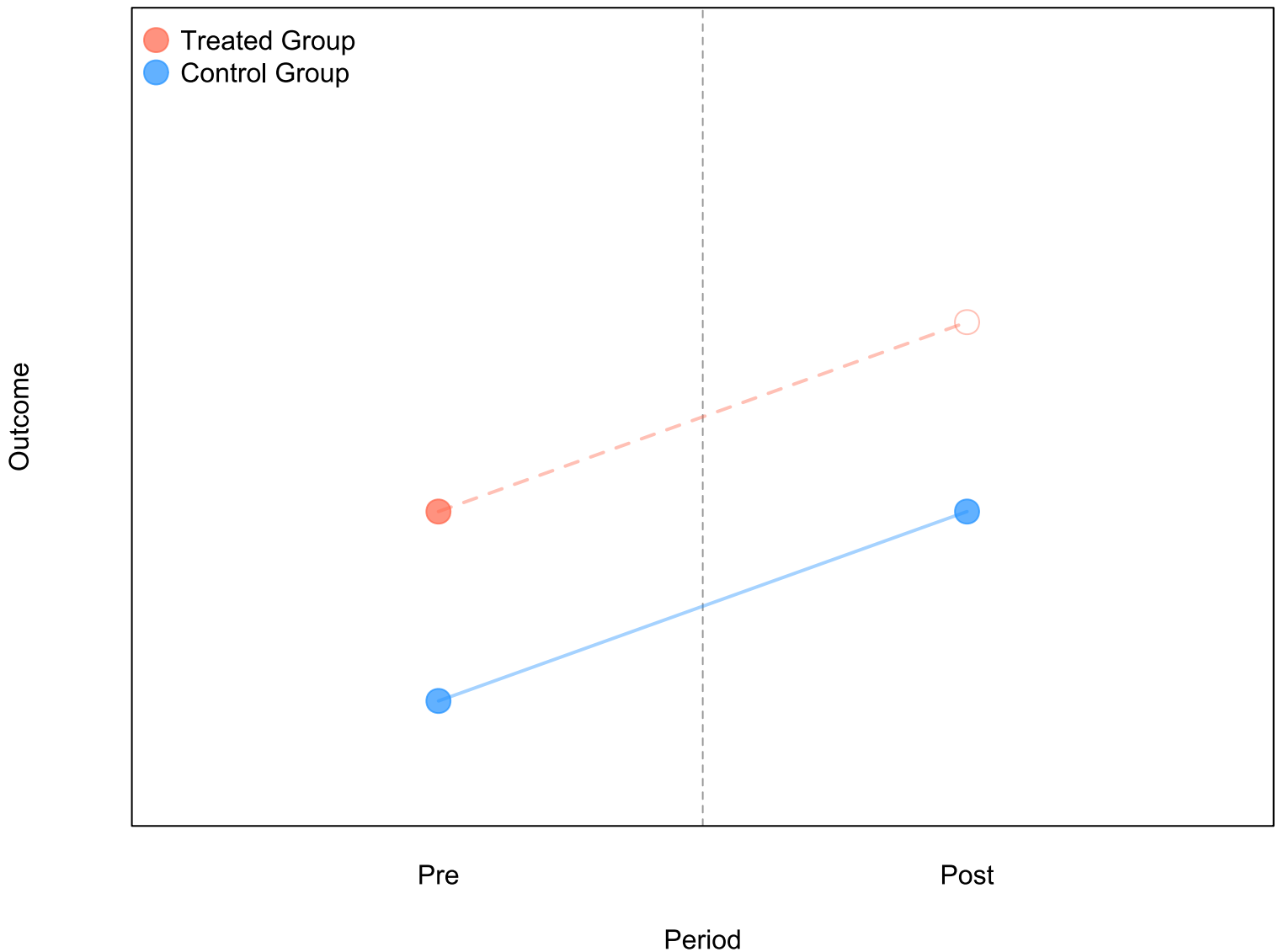
What is Difference-in-Differences?



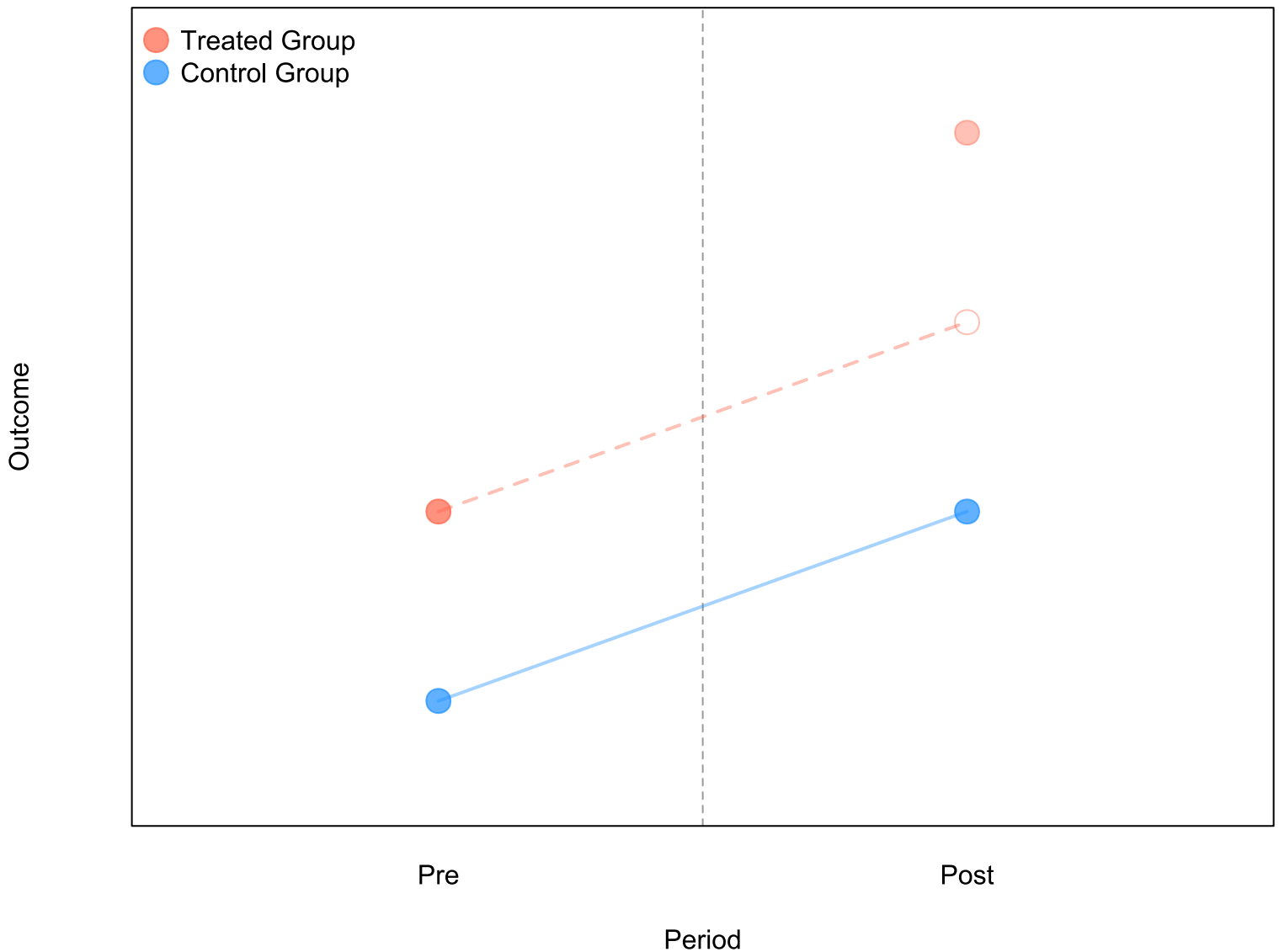
What is Difference-in-Differences?



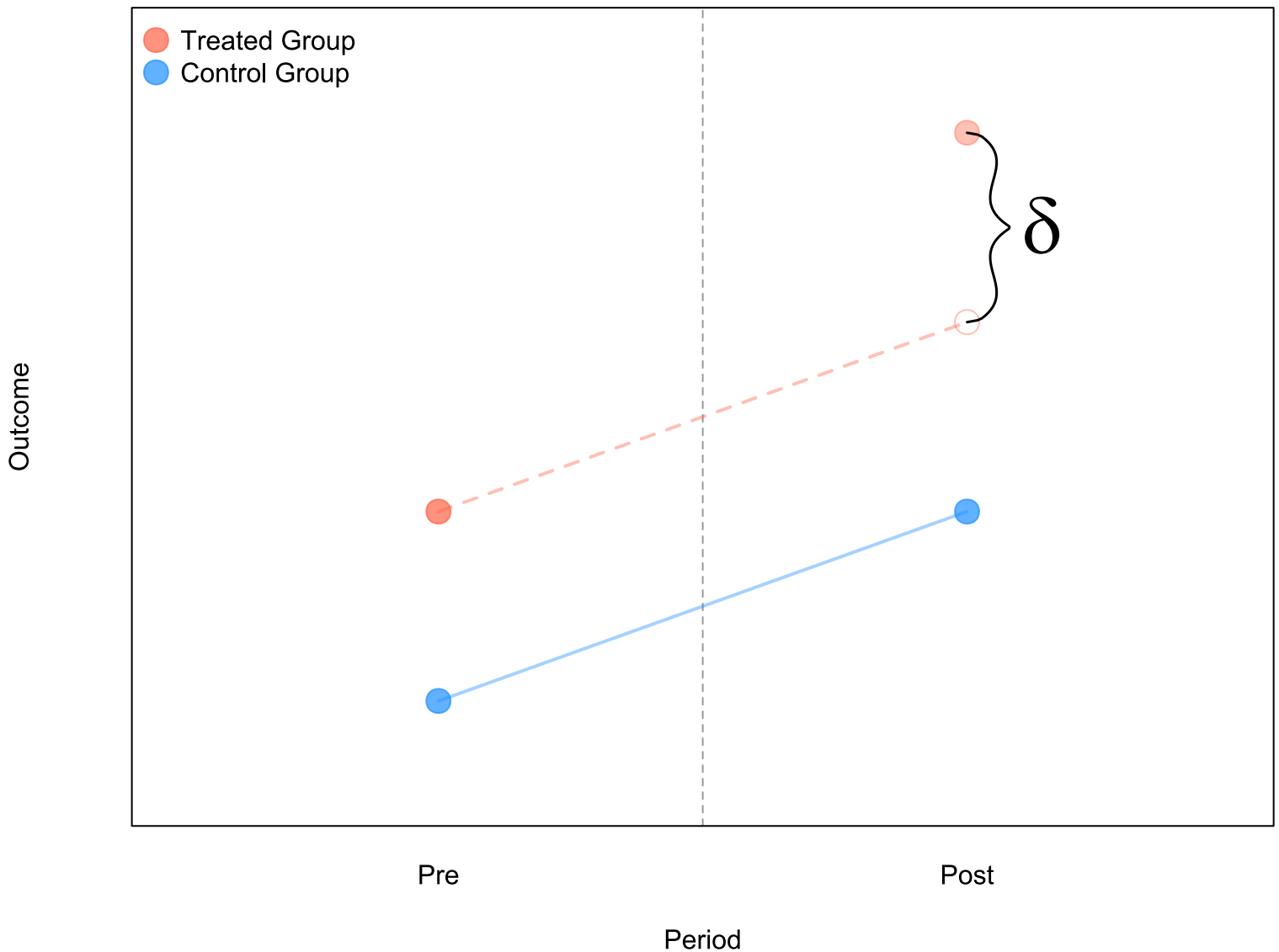
What is Difference-in-Differences?



What is Difference-in-Differences?



What is Difference-in-Differences?

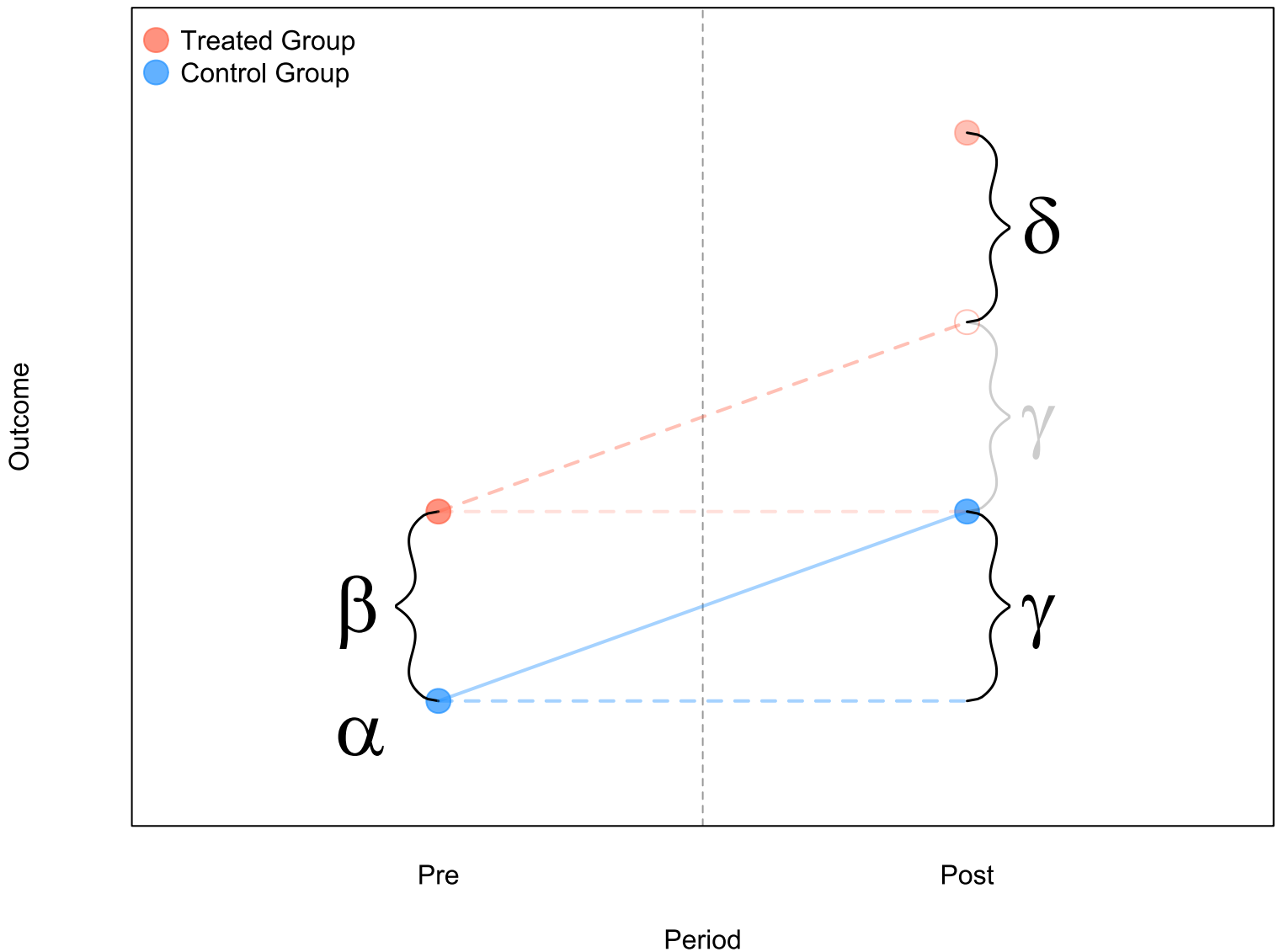


How to use DiD

$$Y_{st} = \alpha + \beta \text{Treat}_s + \gamma \text{Post}_t + \delta(\text{Treat}_s \times \text{Post}_t) + \epsilon_{st}$$

- α represents the pre-period average for the control group.
- β represents initial difference between treatment and control groups.
- γ represents the change in the control group post-treatment
- δ represents the *additional* change, on top of γ, for the treatment group.

How to use DiD

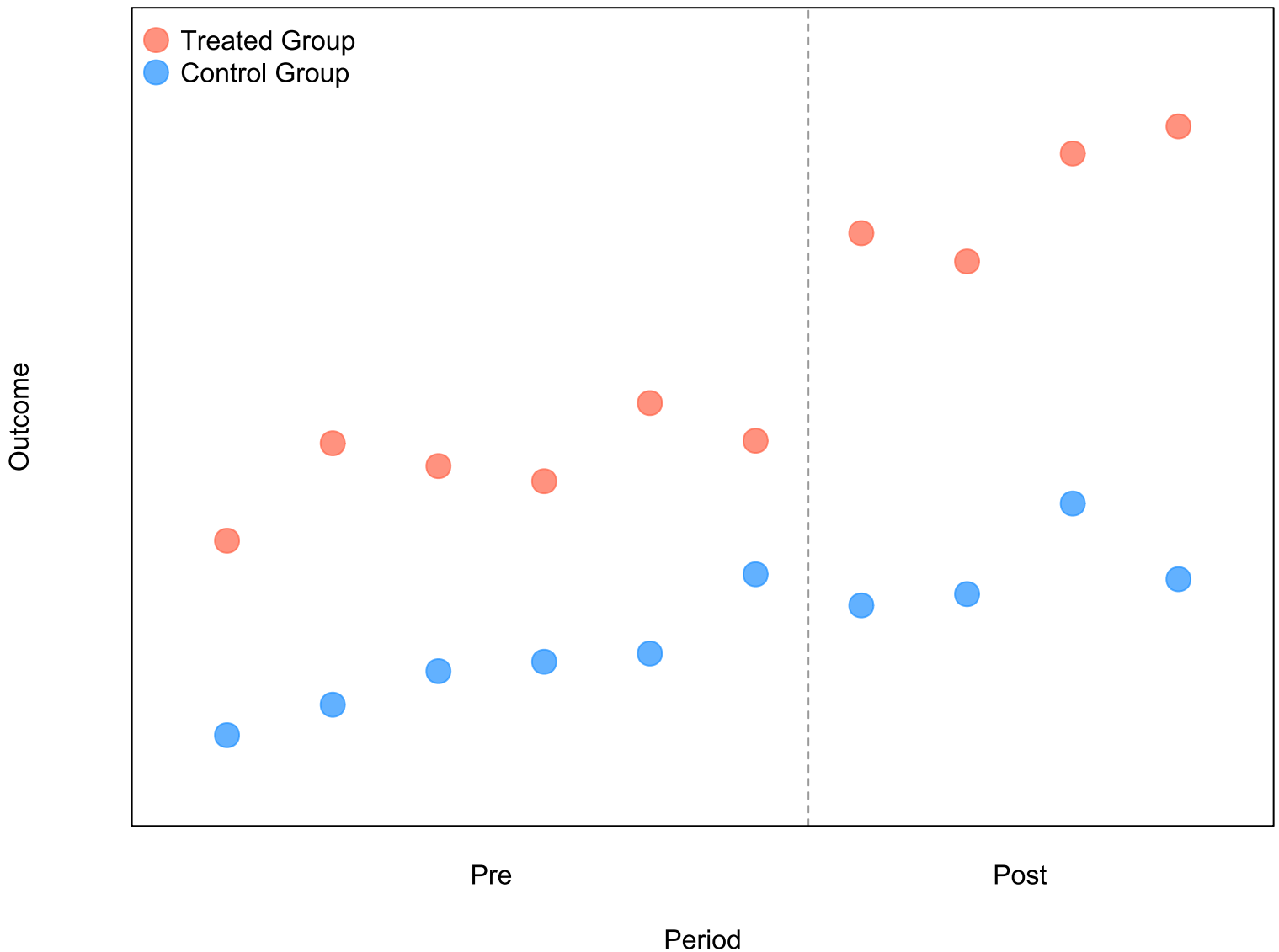


How to use DiD

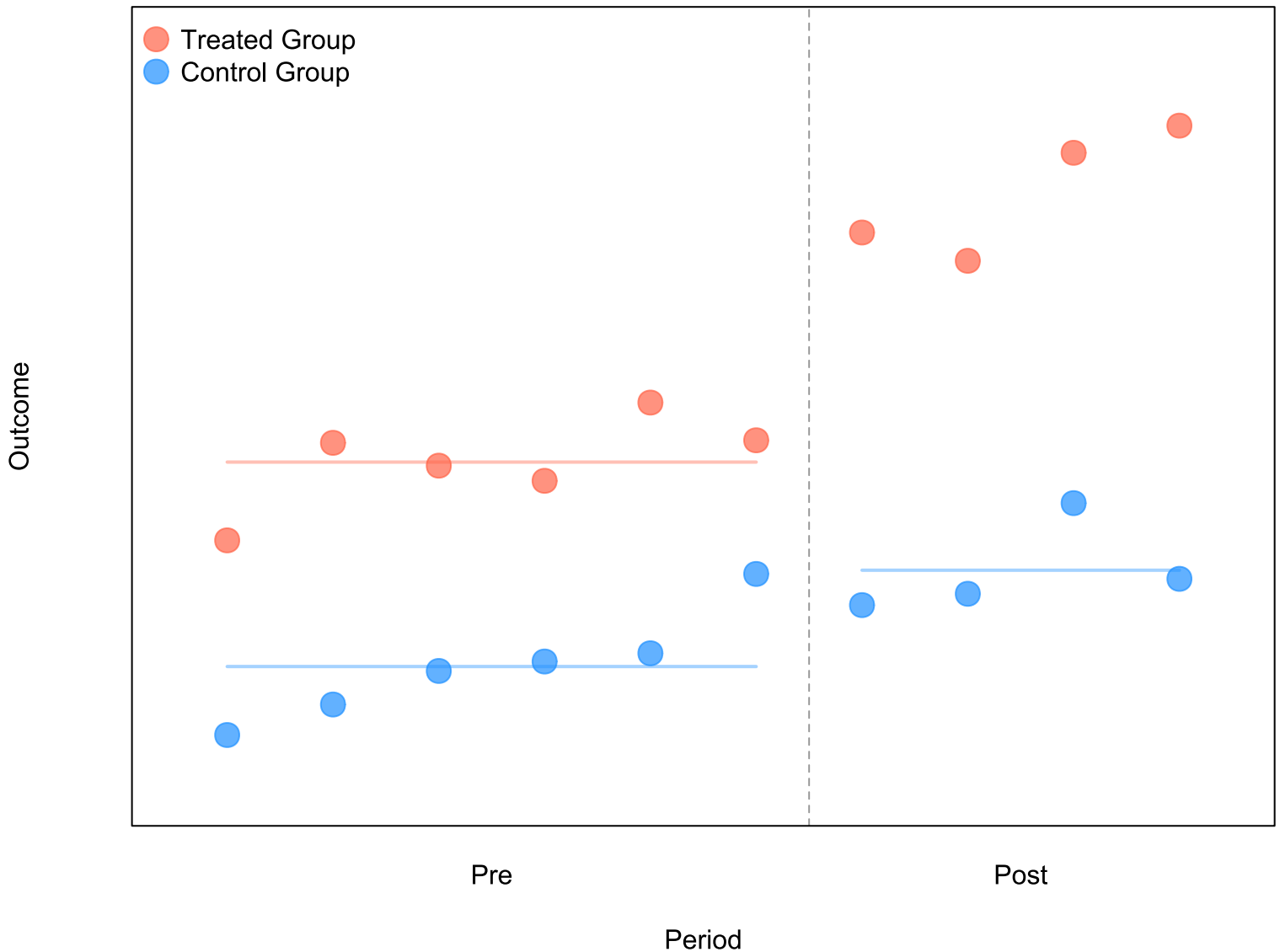
Under what conditions does this work?

1. We must assume that the treatment group would have moved in parallel had they not received treatment. This is called the parallel trends assumption. **Note**: this has nothing to do with the units having parallel trends prior to treatment.
2. We have to assume that treatment is unanticipated, otherwise we cannot attribute the change to the treatment as opposed to preemptive actions or expectations.
3. Stable Unit Treatment Value Assumption (SUTVA): treatment does not spillover onto units in the control group.

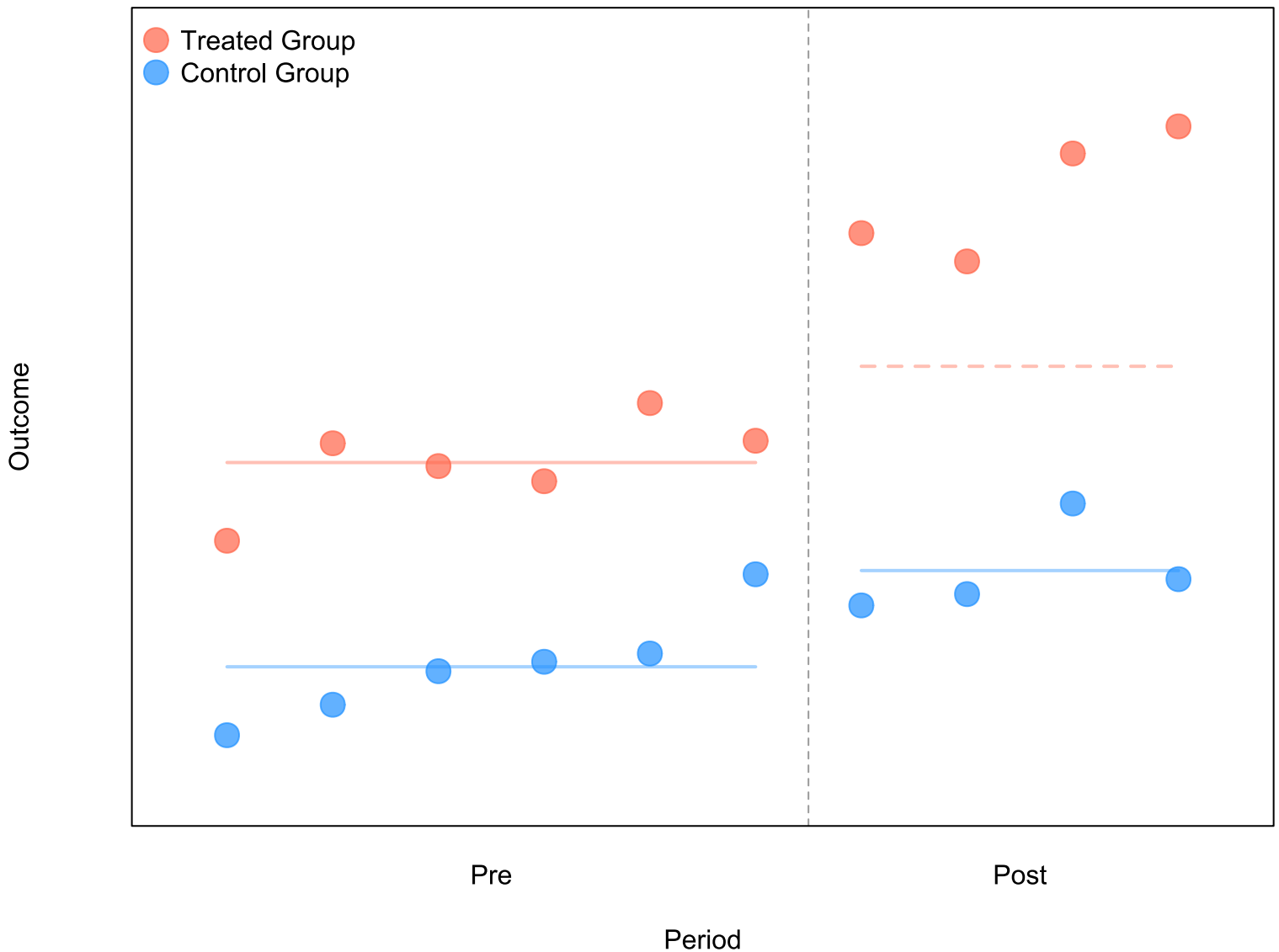
Two-Way Fixed Effects



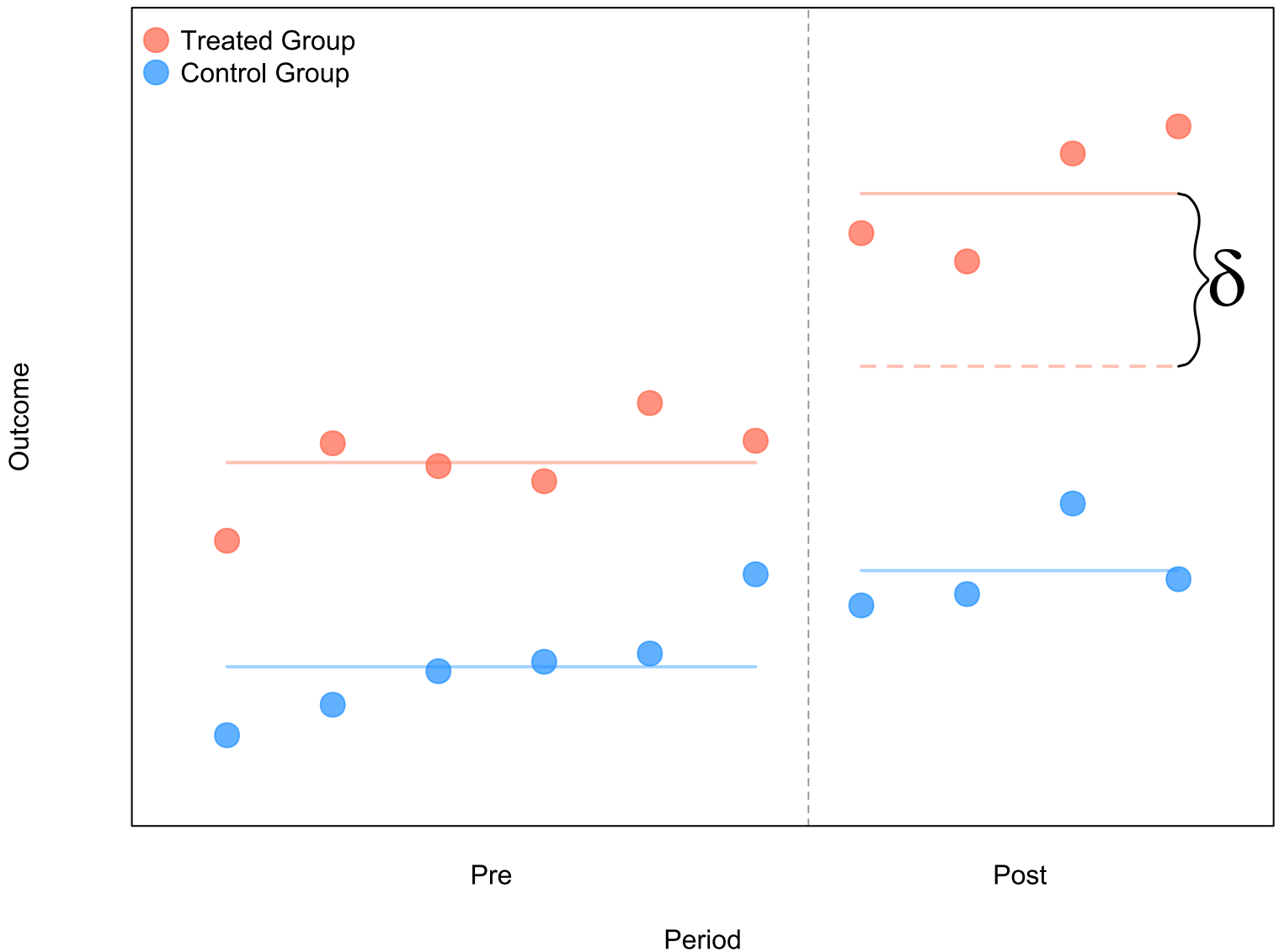
Two-Way Fixed Effects



Two-Way Fixed Effects



Two-Way Fixed Effects

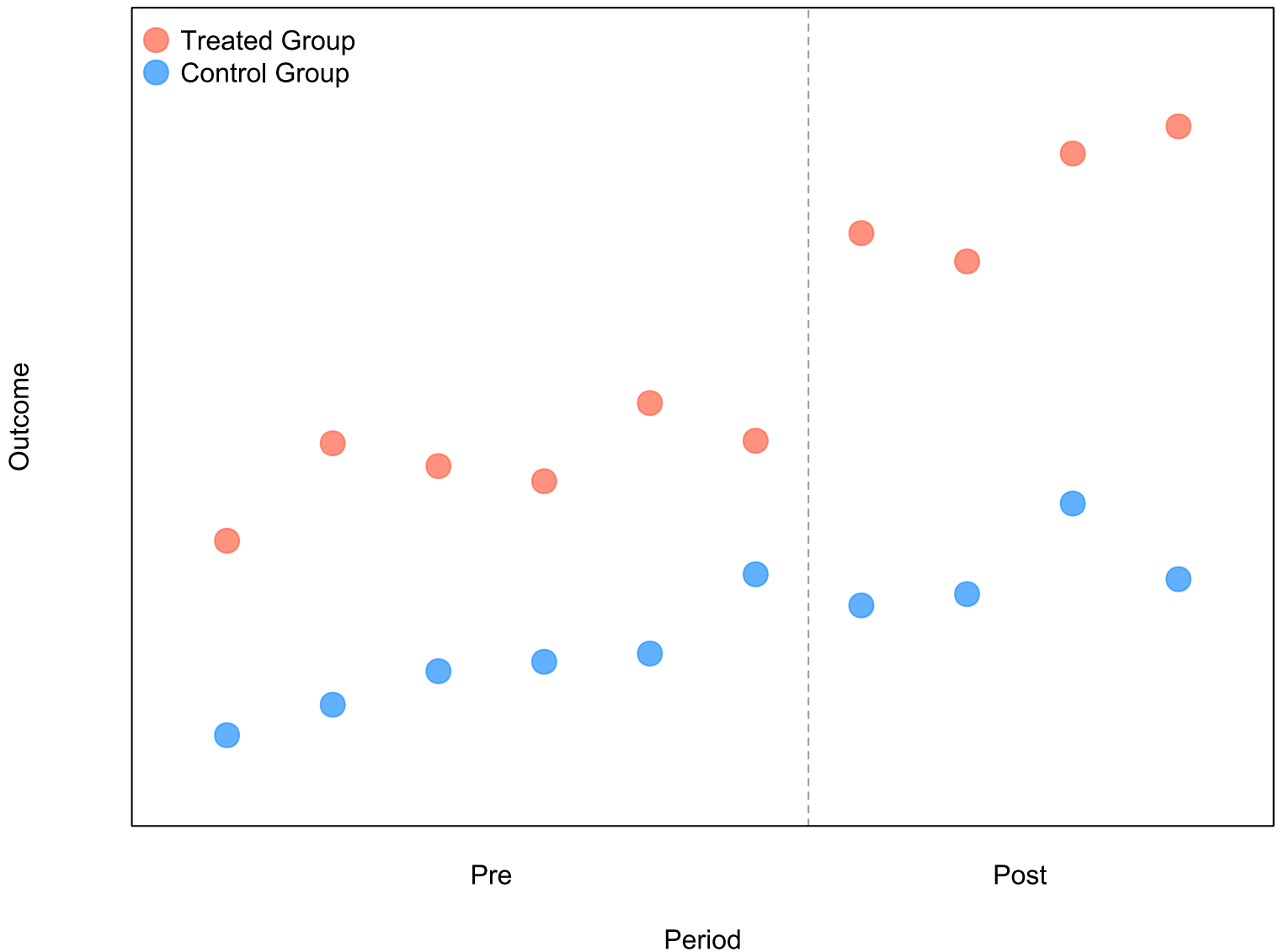


Two-Way Fixed Effects

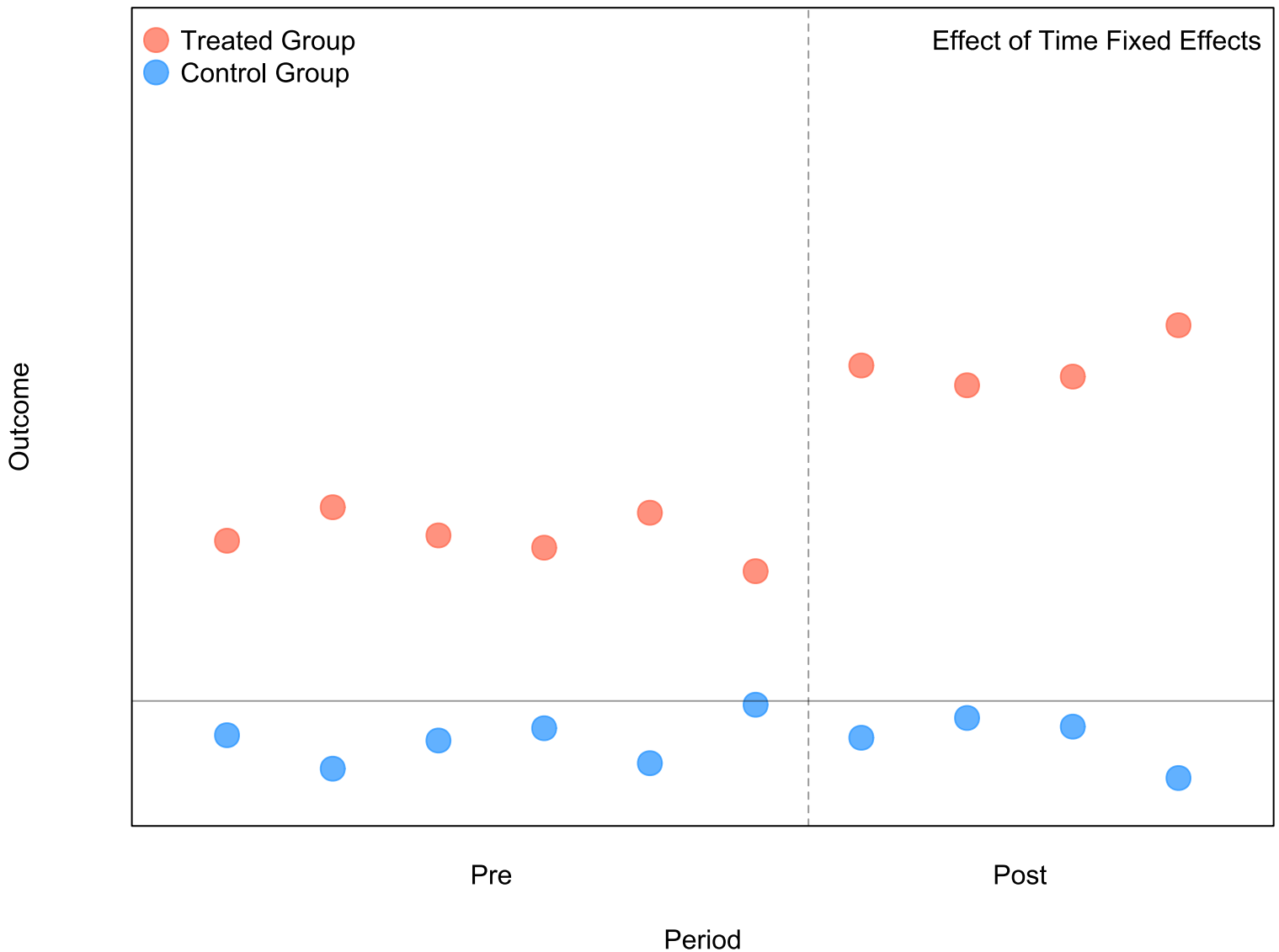
So far, we have assumed that the treatment and control groups can easily be represented by single averages, though this is not a necessary assumption. We can build in additional flexibility by using "fixed effects" in our model:

$$
\begin{aligned}
Y_{st} &= \alpha + \beta \text{Treat}_s + \gamma \text{Post}_t + \delta(\text{Treat}_s \times \text{Post}_t) + \epsilon_{st} \\
&= \beta_s + \gamma \text{Post}_t + \delta(\text{Treat}_s \times \text{Post}_t) + \epsilon_{st} \\
&= \beta_s + \gamma_t + \delta(\text{Treat}_s \times \text{Post}_t) + \epsilon_{st}
\end{aligned}
$$

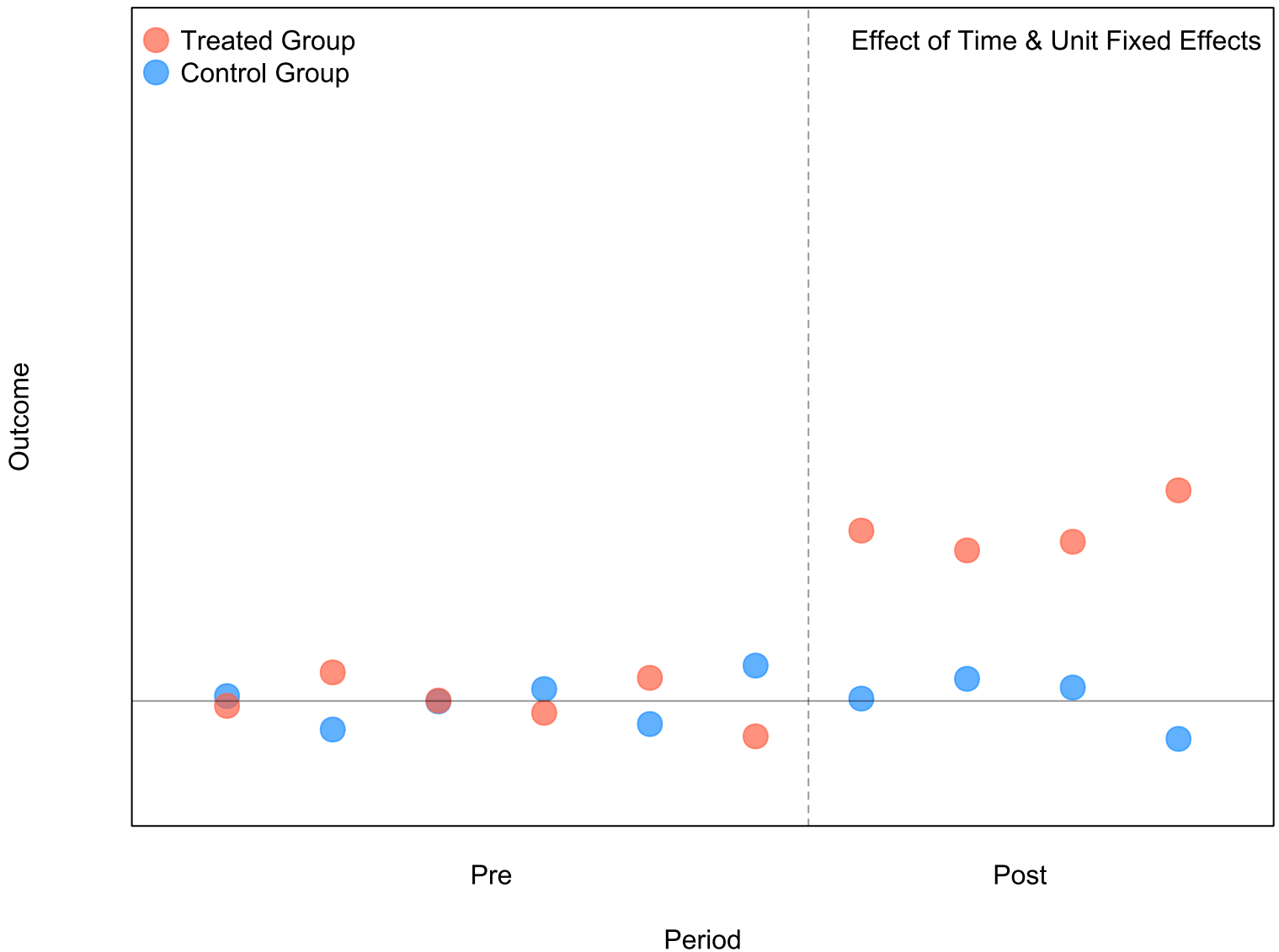
Two-Way Fixed Effects



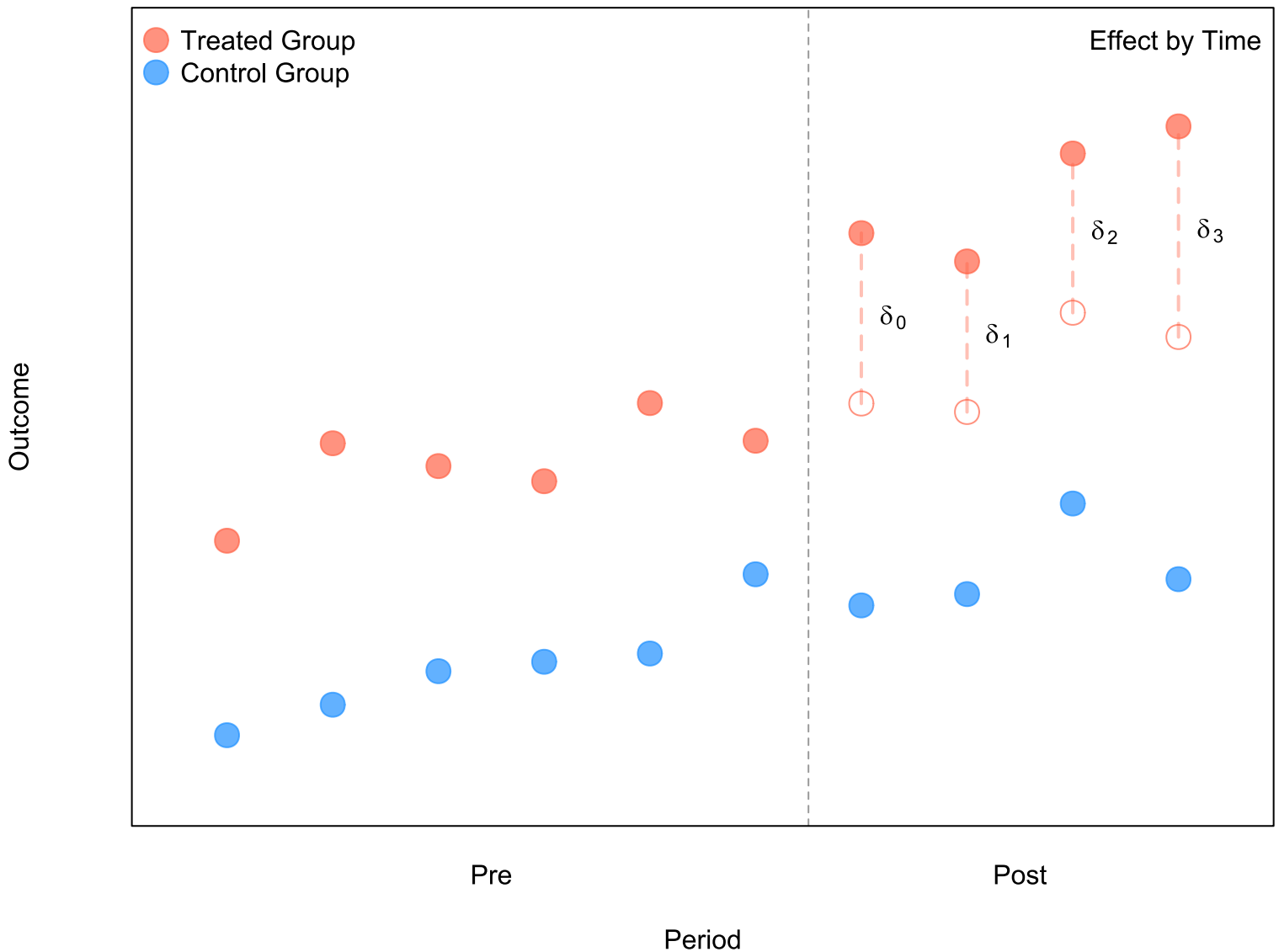
Two-Way Fixed Effects



Two-Way Fixed Effects



Two-Way Fixed Effects



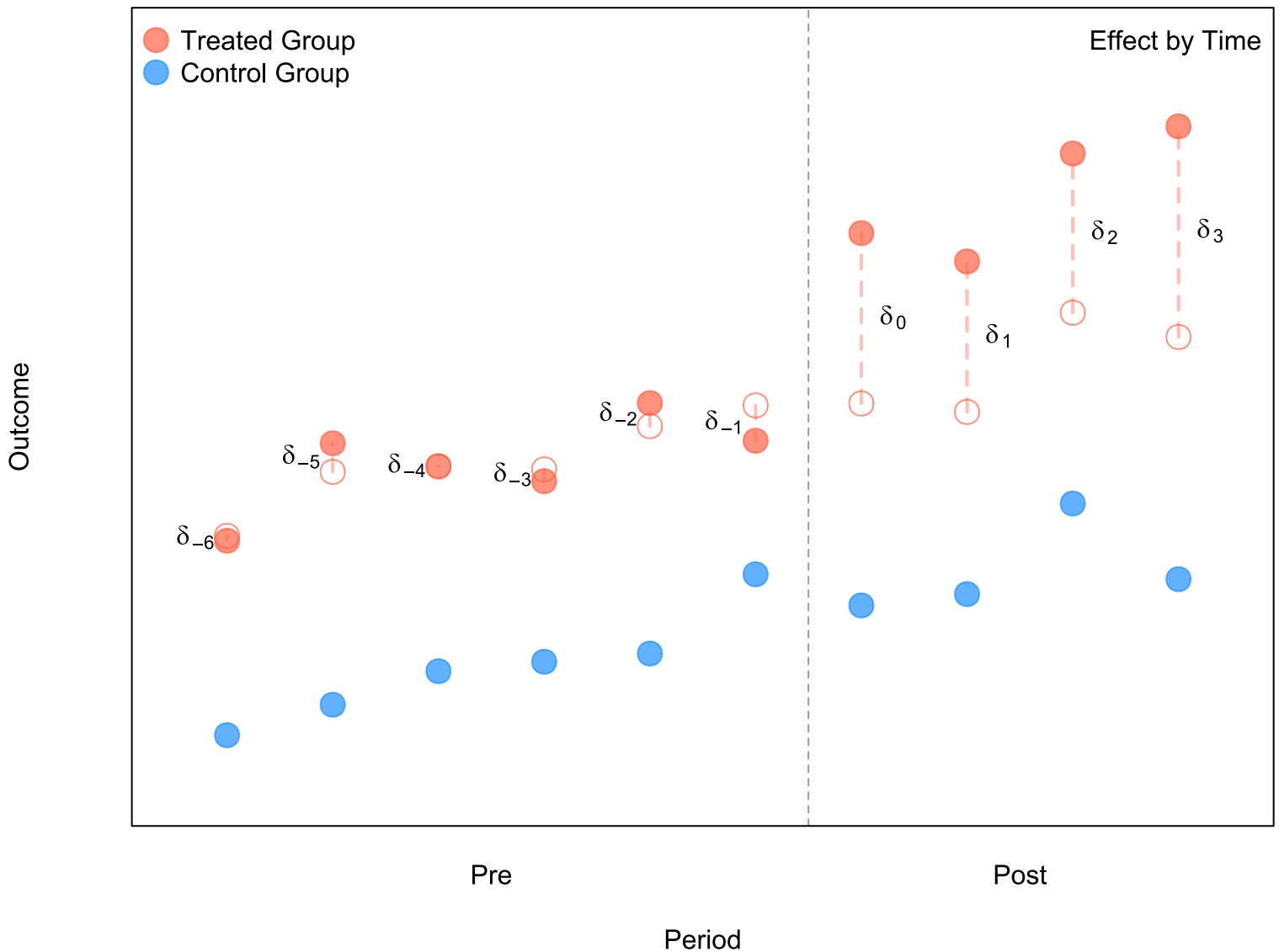
Event Studies

Once we have $\hat{\delta}_0, \ldots, \hat{\delta}_n$, OLS can aggregate them into $\overline{\delta}$, which represents the average treatment effect.

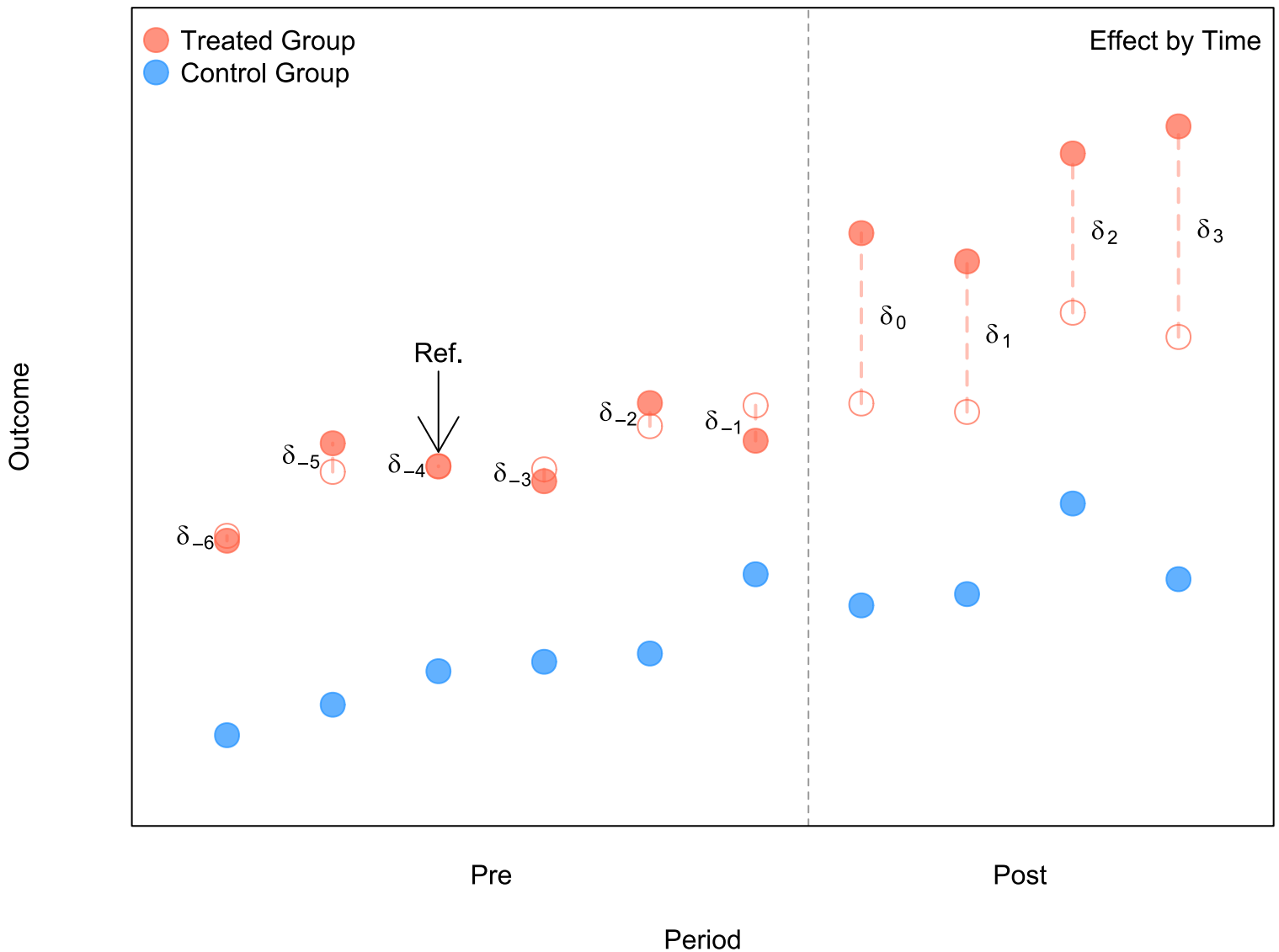
However, note that if we do *not* aggregate, we get estimates that correspond to the dynamic nature of the treatment effect. We can even get estimates leading up to the treatment to check for anticipation or a lack of parallel pre-trends.

Economists often refer to dynamic DiD estimates as *Event Studies*. To avoid perfect multicolinearity, researchers need to select a single period to "omit" or use as reference. Researchers generally choose the time period immediately prior to treatment.

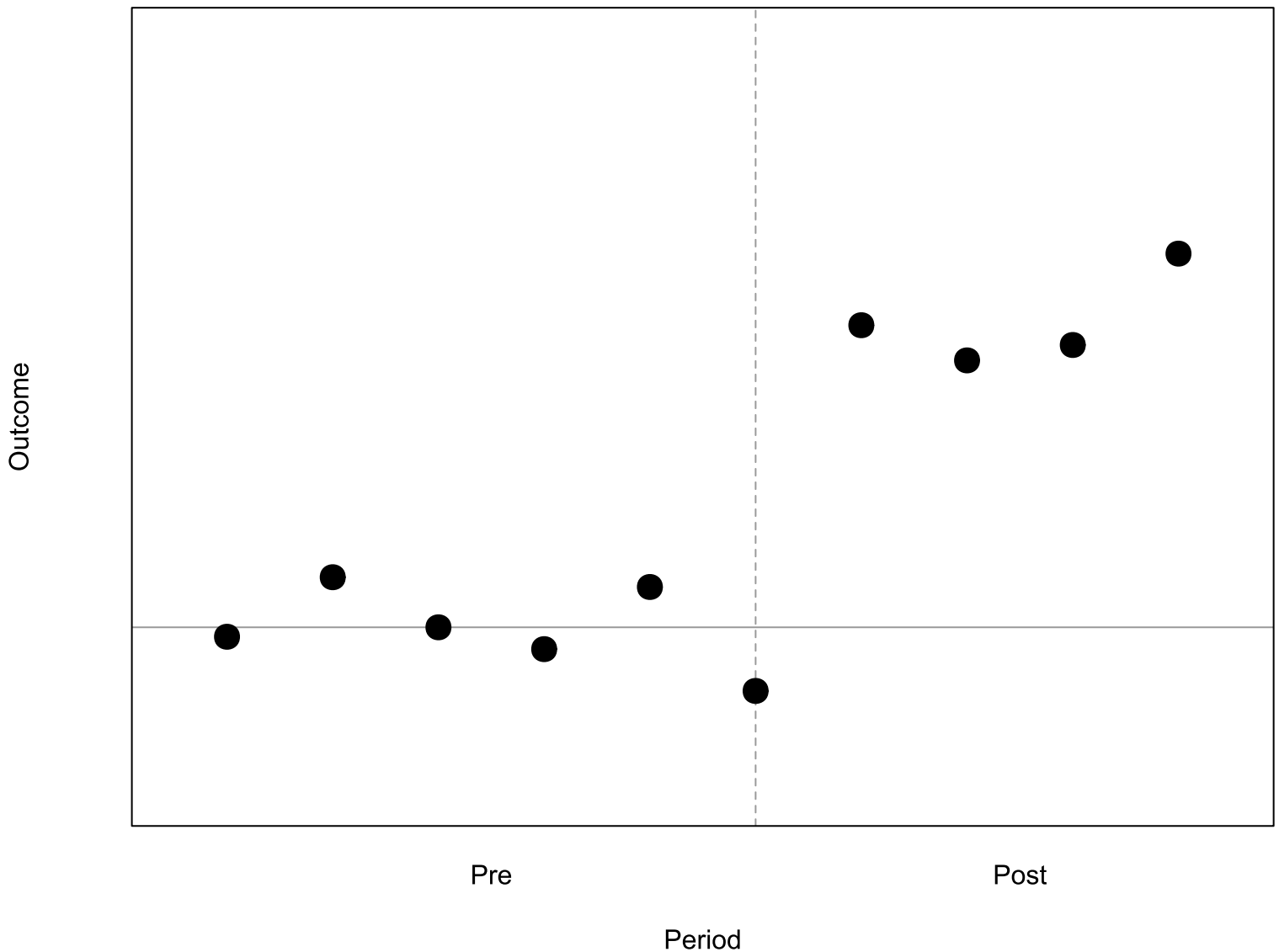
Event Studies



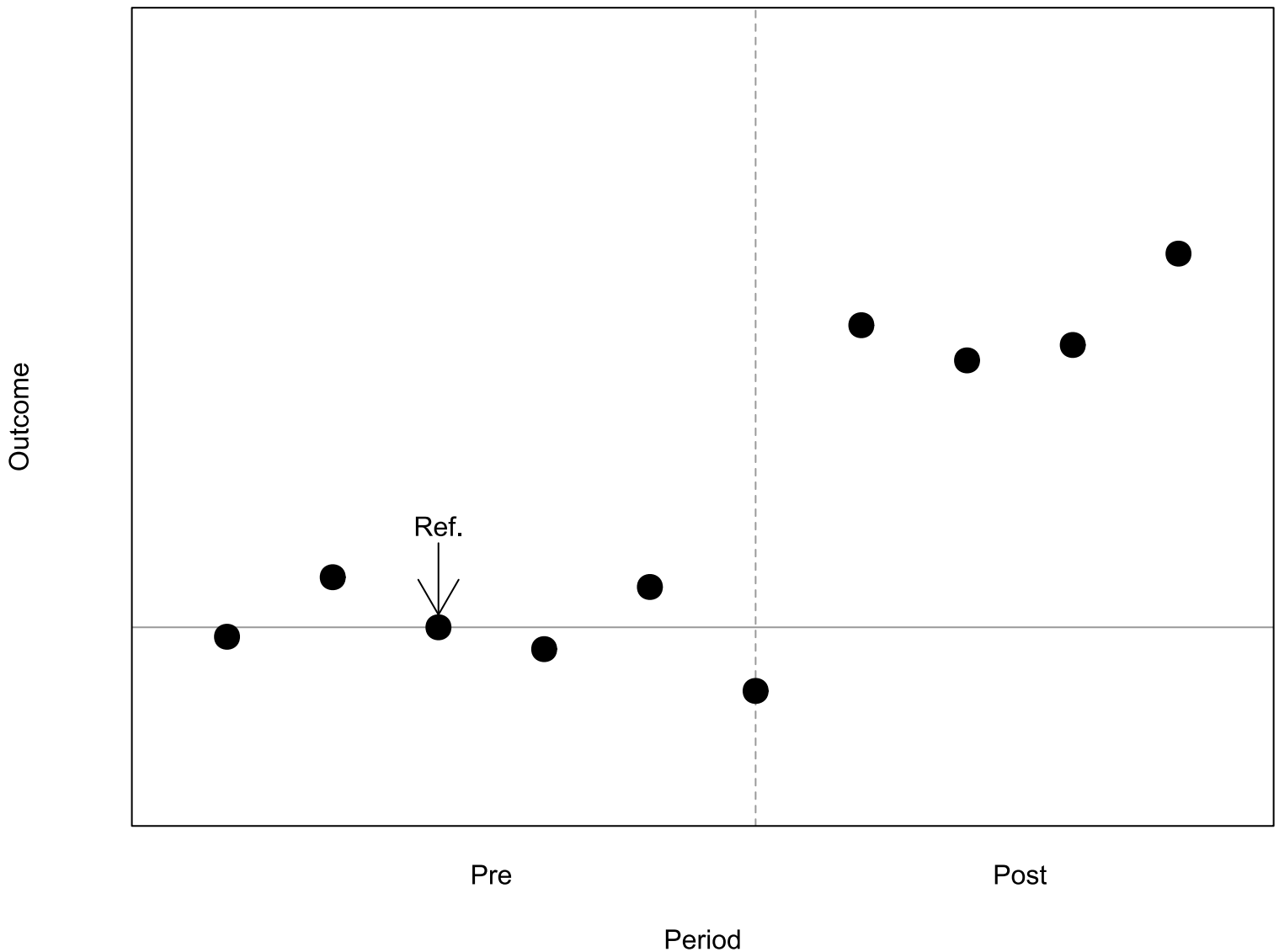
Event Studies



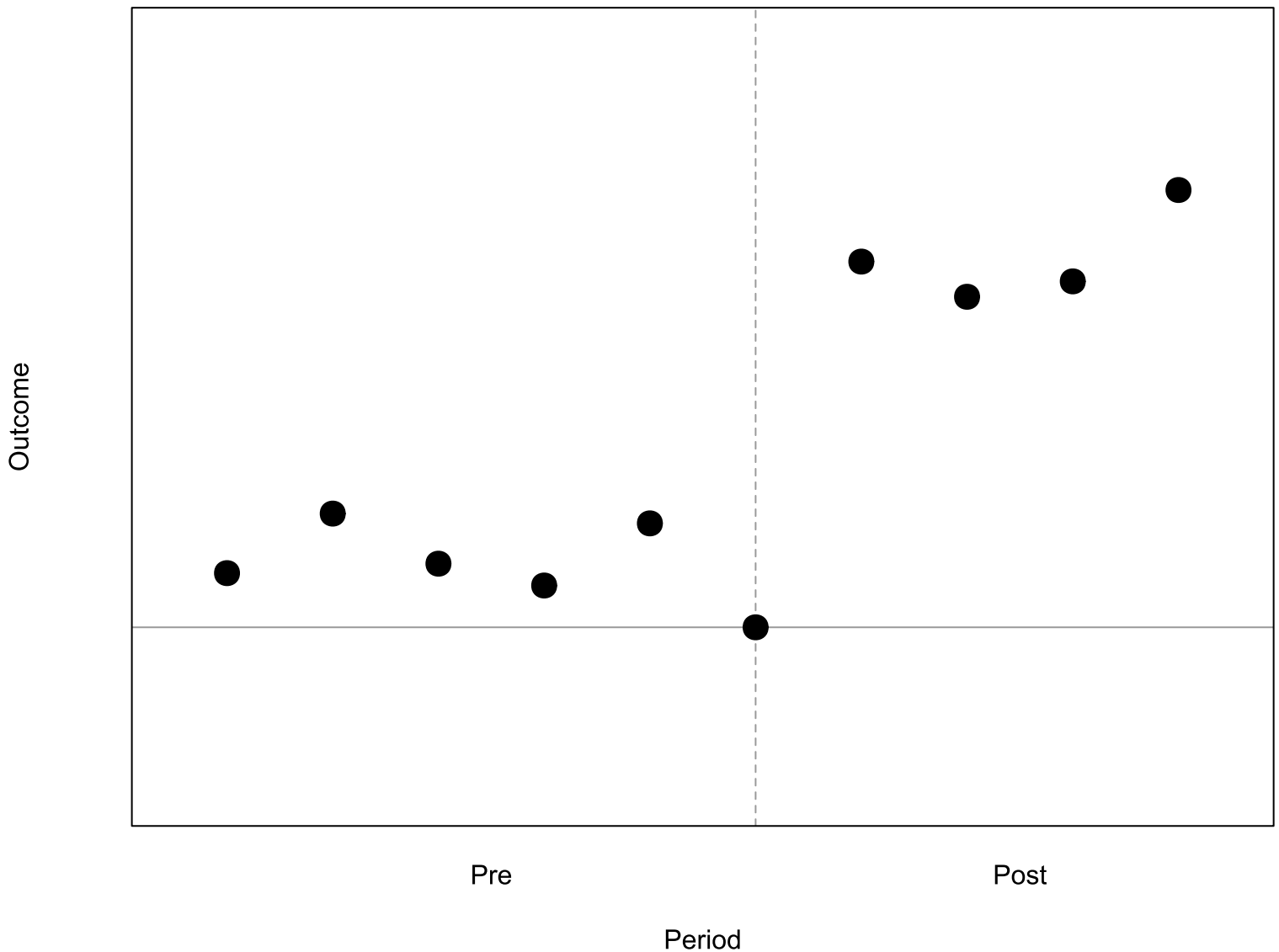
Event Studies



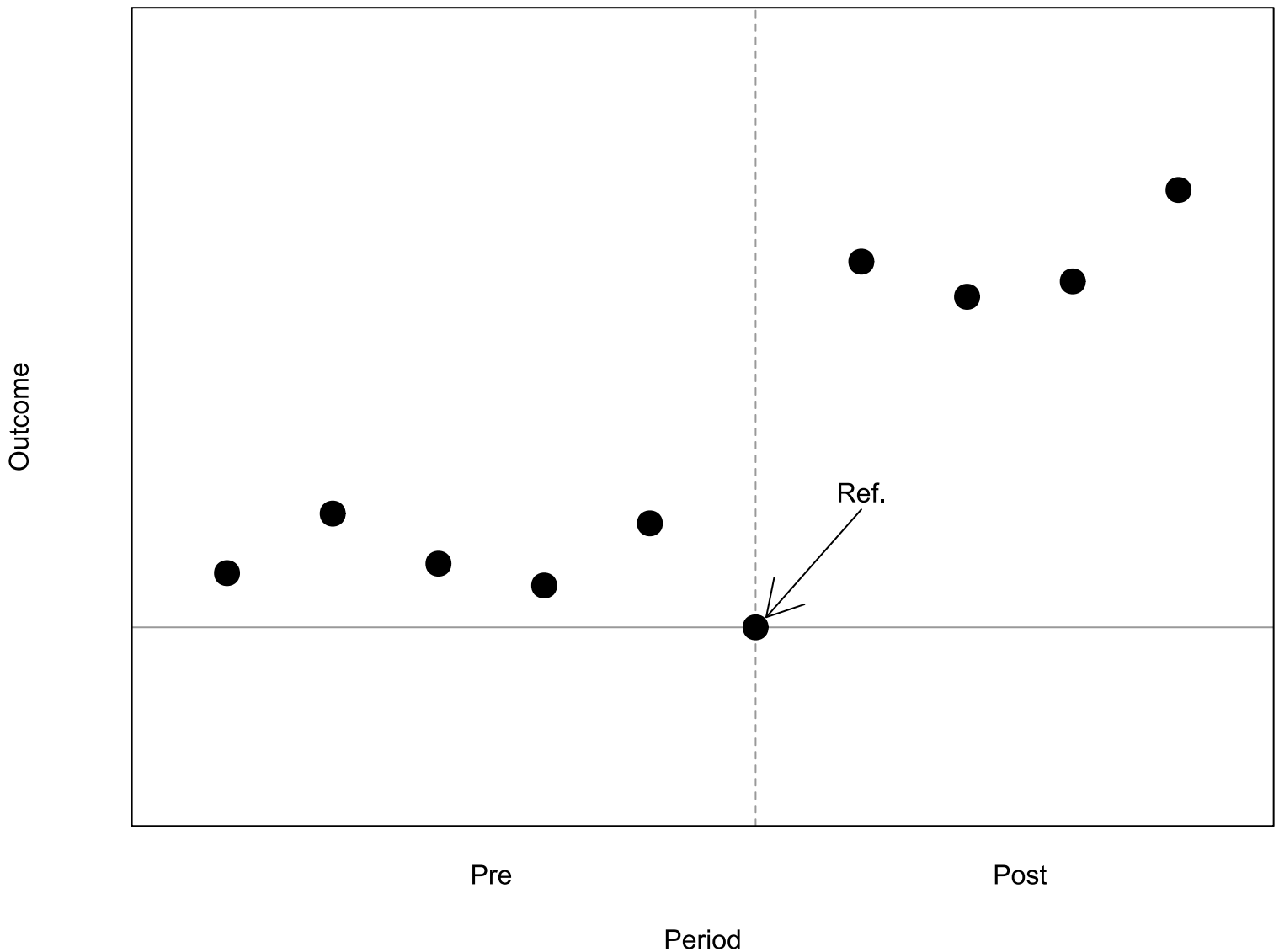
Event Studies



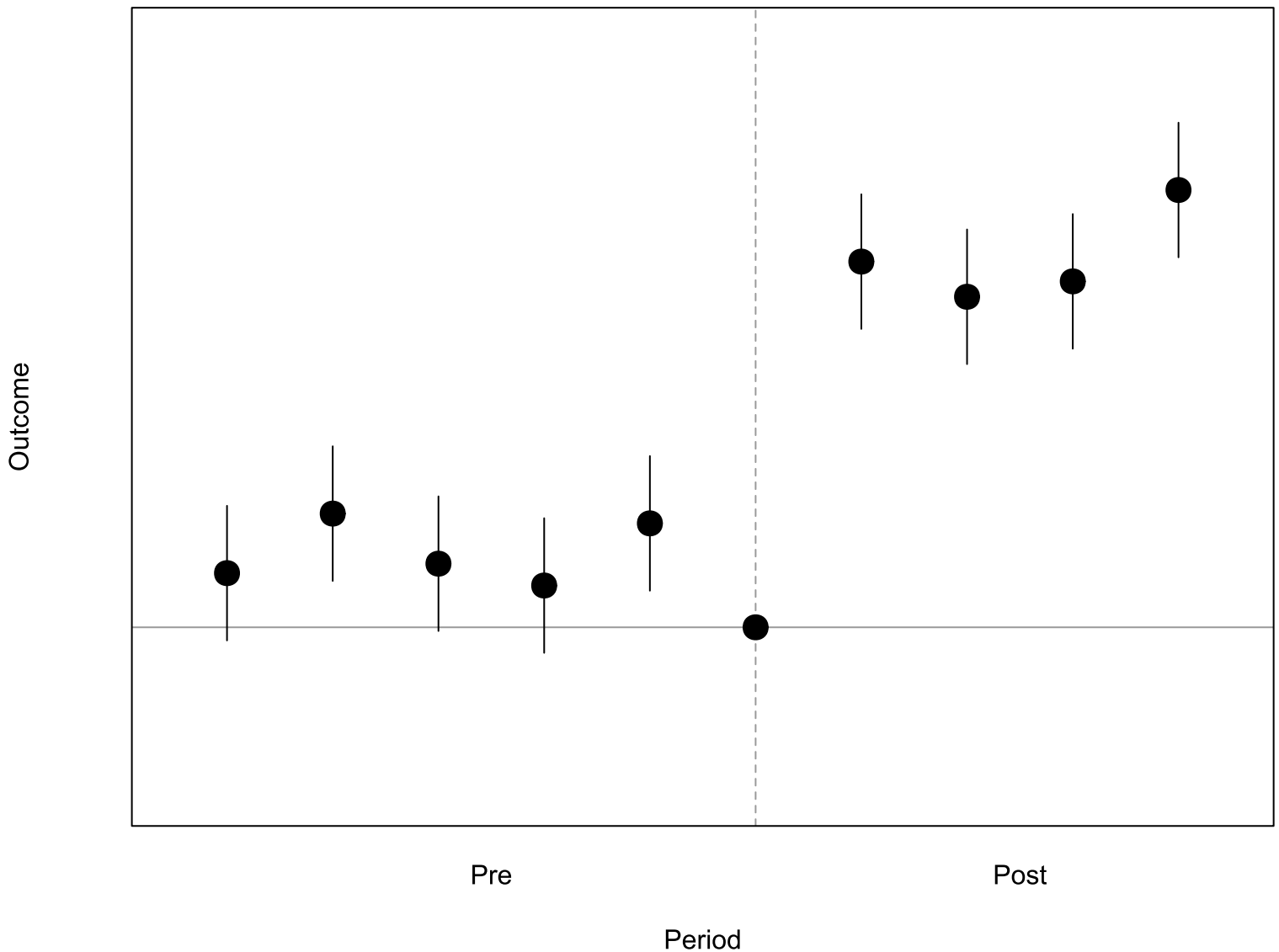
Event Studies



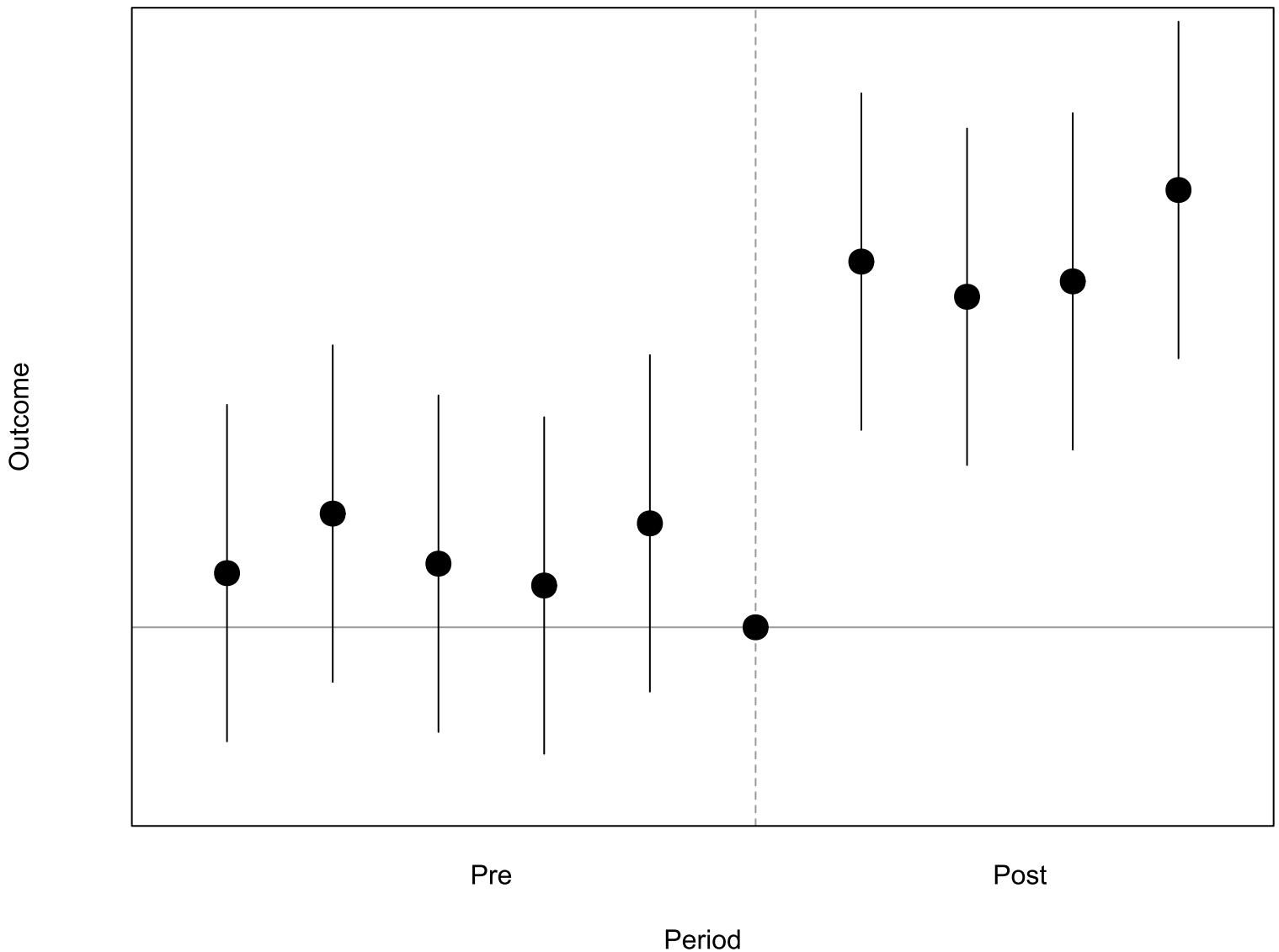
Event Studies



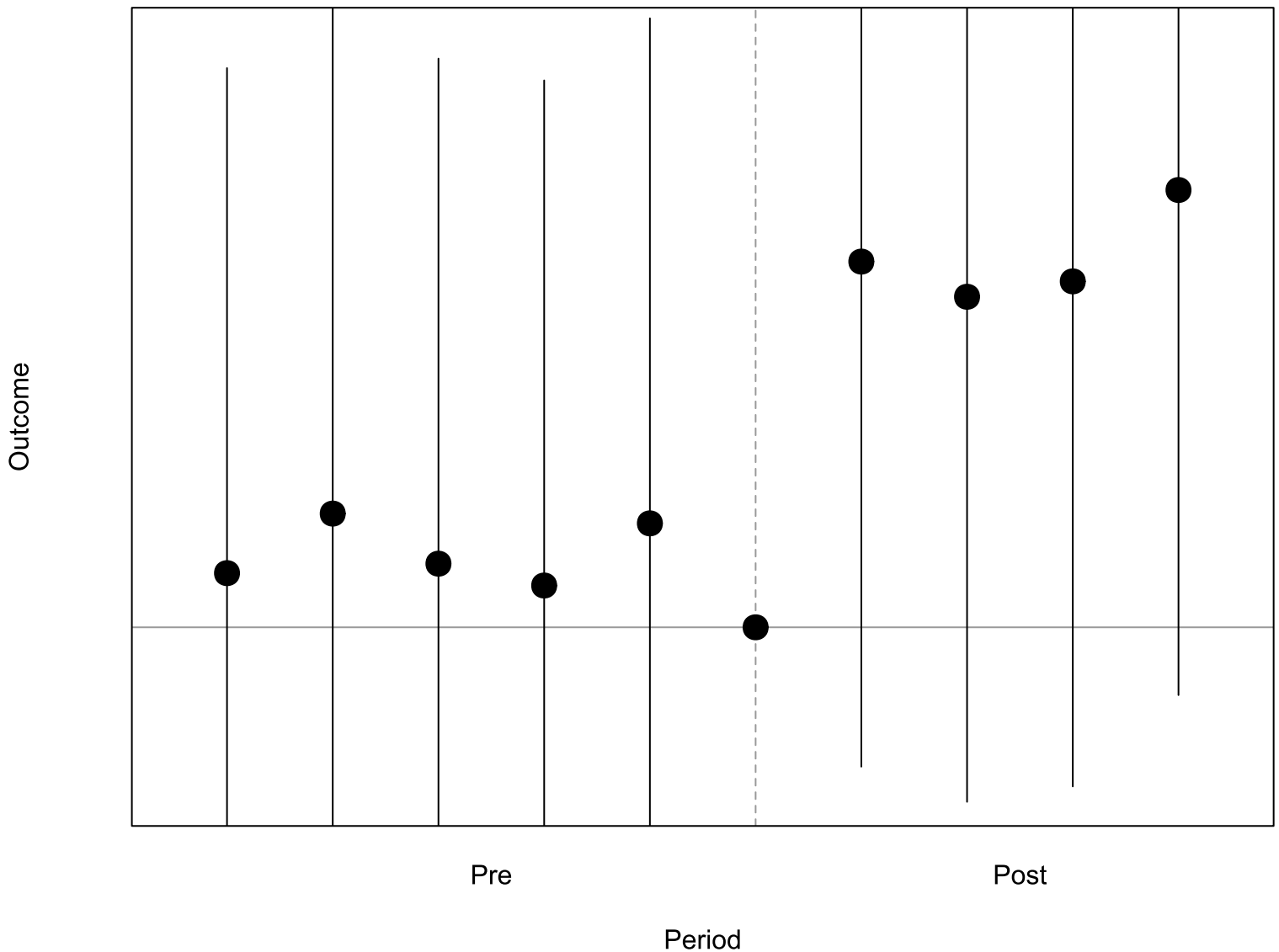
Event Studies



Event Studies



Event Studies



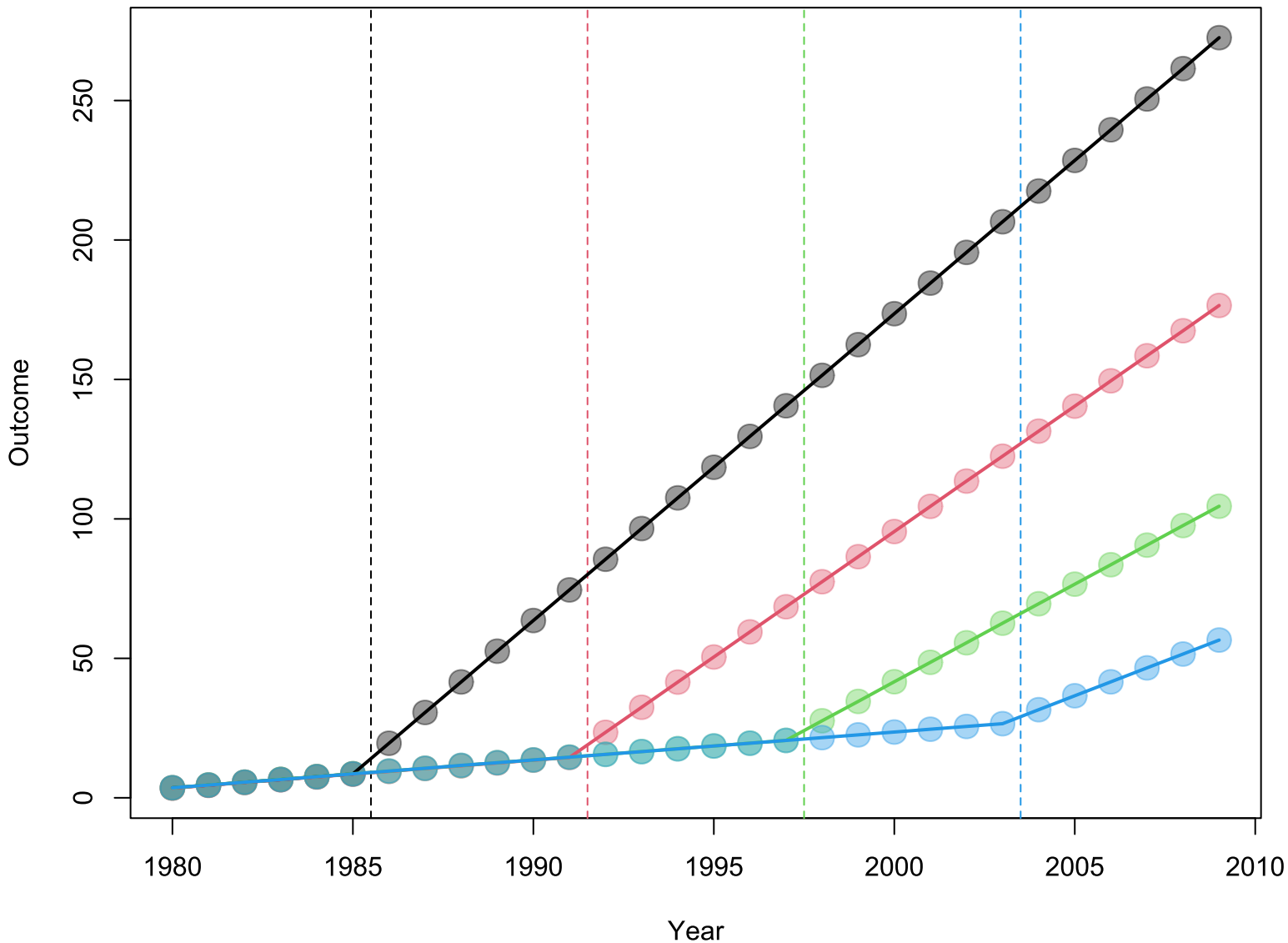
Staggered Treatment Timing

Fixed Effects and Event Studies are especially helpful when studying settings with heterogeneous/dynamic treatment effects or staggered treatment timing (e.g., regional law, or regulation mandates, program rollouts, firm policy adoptions, etc.).

When treatment is staggered, there is no one single treatment / control group and there is no single pre / post period. There is a different post period for each unit and a different treatment group composition in each period. TWFE can "easily" handle this by allowing each unit and period to have its own estimate.

However, econometricians have found TWFE to occasionally be problematic when treatment is staggered and heterogeneous.

Staggered Treatment Timing



Staggered Treatment Timing

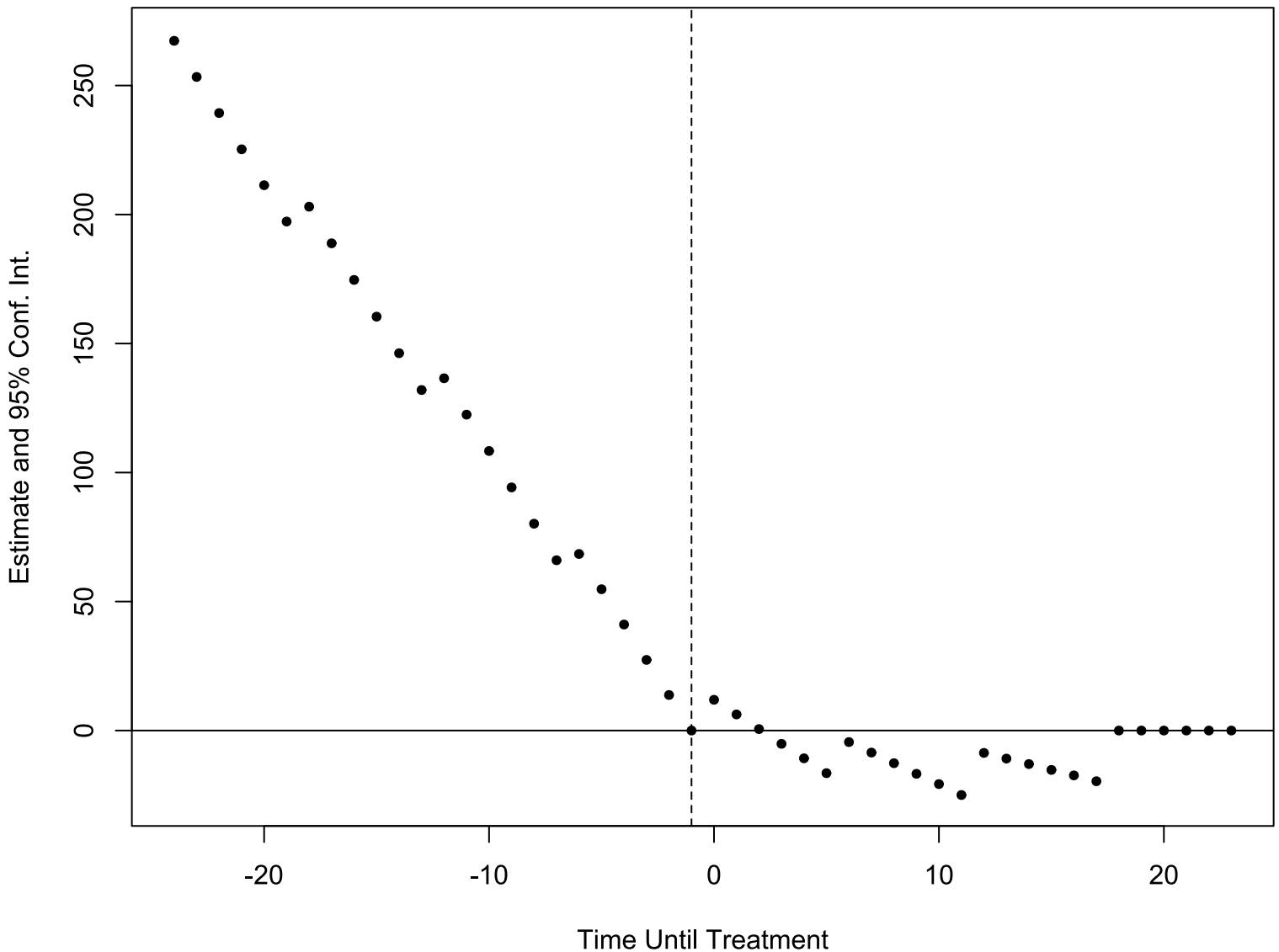
These data contain 1,000 "units" (index by s) split equally into four groups and observed over 30 "years" (indexed by t). The TWFE estimation:

$$Y_{st} = \beta_s + \gamma_t + \delta(\text{Treat}_s \times \text{Post}_t) + \epsilon_{st}$$

▶ Code

	(1)
Treat x Post	-6.690***
	(0.670)
Num.Obs.	30000

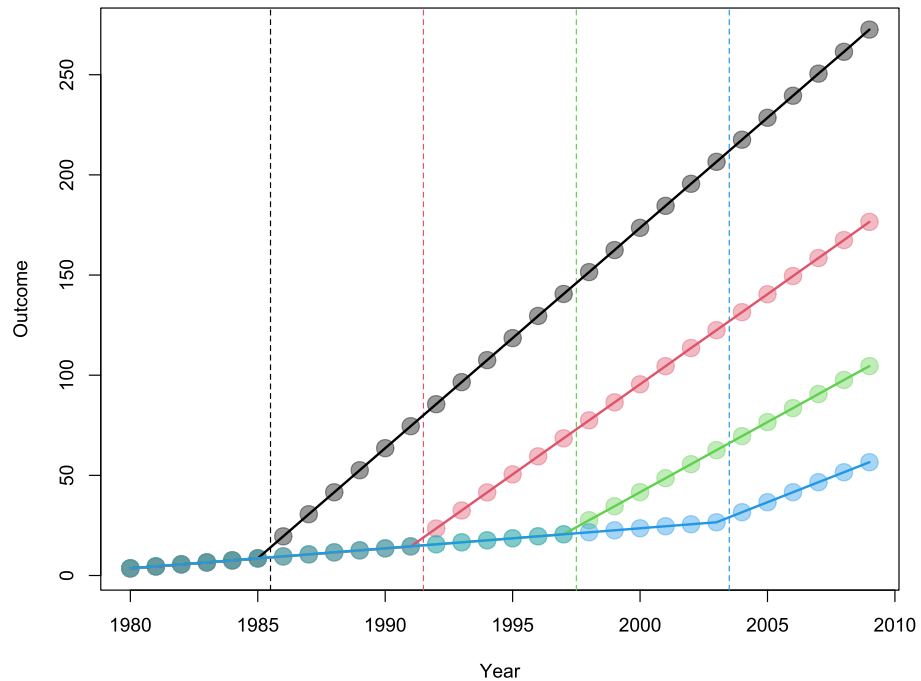
Staggered Treatment Timing



Staggered Treatment Timing

When treatment timing is staggered, we have four possible groups to consider:

1. Always treated
2. Earlier Treated
3. Later Treated
4. Never treated



Problem 1: Regression is not smart enough to know that we do *not* want it to use already-treated units when constructing counterfactuals of newly-treated units.

Problem 2: Treatment heterogeneity contaminates FEs depending how long a unit is treated or how many units within a time period are treated.

Staggered Treatment Timing

To diagnose the severity of the problem, we can use a *Goodman-Bacon Decomposition*:

Type	Weight	Estimate
Earlier vs Later Treated	0.50	51.80
Later vs Earlier Treated	0.50	-65.18
Total	1.00	-6.69

“Fixing” the Problem

There are two approaches we can use:

1. Eliminate inappropriate comparisons.
2. Get unbiased estimates of our fixed effect.

Today, I’ll briefly demonstrate the latter using *Two-Stage Difference-in-Differences*.

Two-Stage DiD

$$Y_{st} = \beta_s + \gamma_t + \delta(\text{Treat}_s \times \text{Post}_t) + \epsilon_{st}$$

We need to get $\widehat{\beta}_s, \widehat{\gamma}_t$ from the equation. Note that β_s is time invariant and γ_t is unit invariant, so it shouldn't matter which years we use to estimate β_s (or which units we use to estimate γ_t).

We are therefore going to estimate:

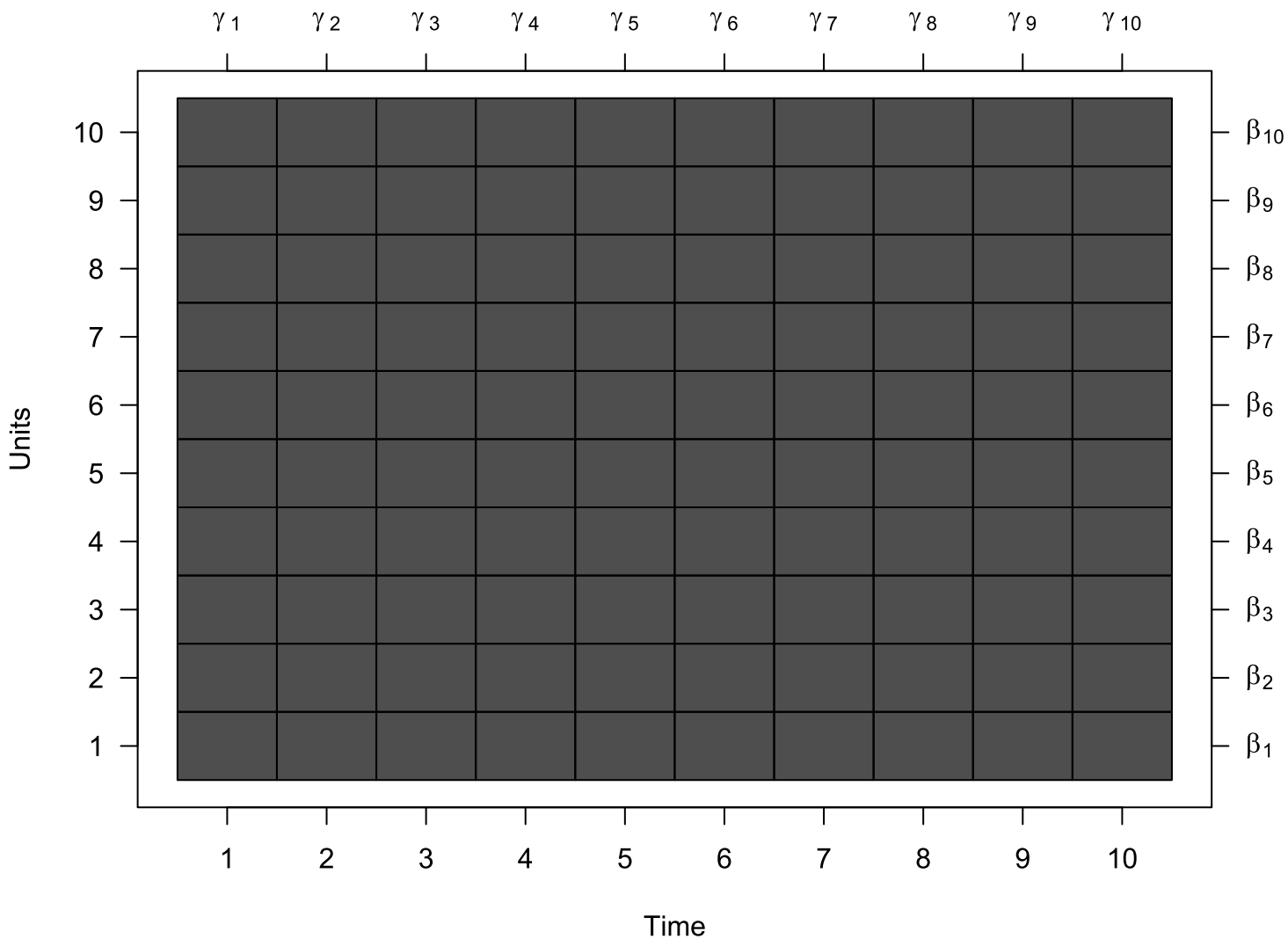
$$Y_{st} = \beta_s + \gamma_t + \epsilon_{st}$$

where $\text{Treat}_s \times \text{Post}_t = 0$.

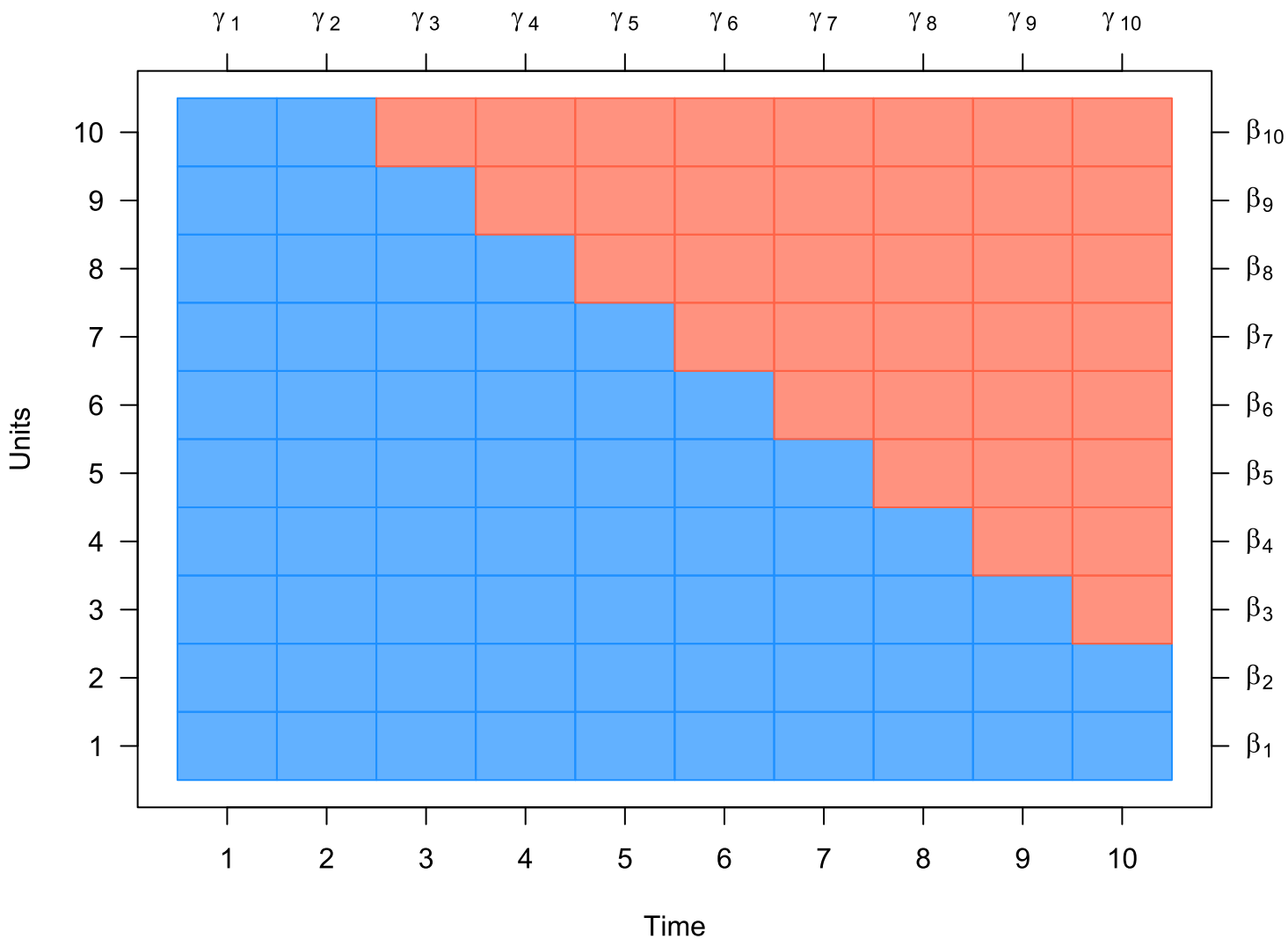
Collect $\widehat{\beta}_s$ and $\widehat{\gamma}_t$, then estimate:

$$Y_{st} - \widehat{\beta}_s - \widehat{\gamma}_t = \delta(\text{Treat}_s \times \text{Post}_t) + \epsilon_{st}$$

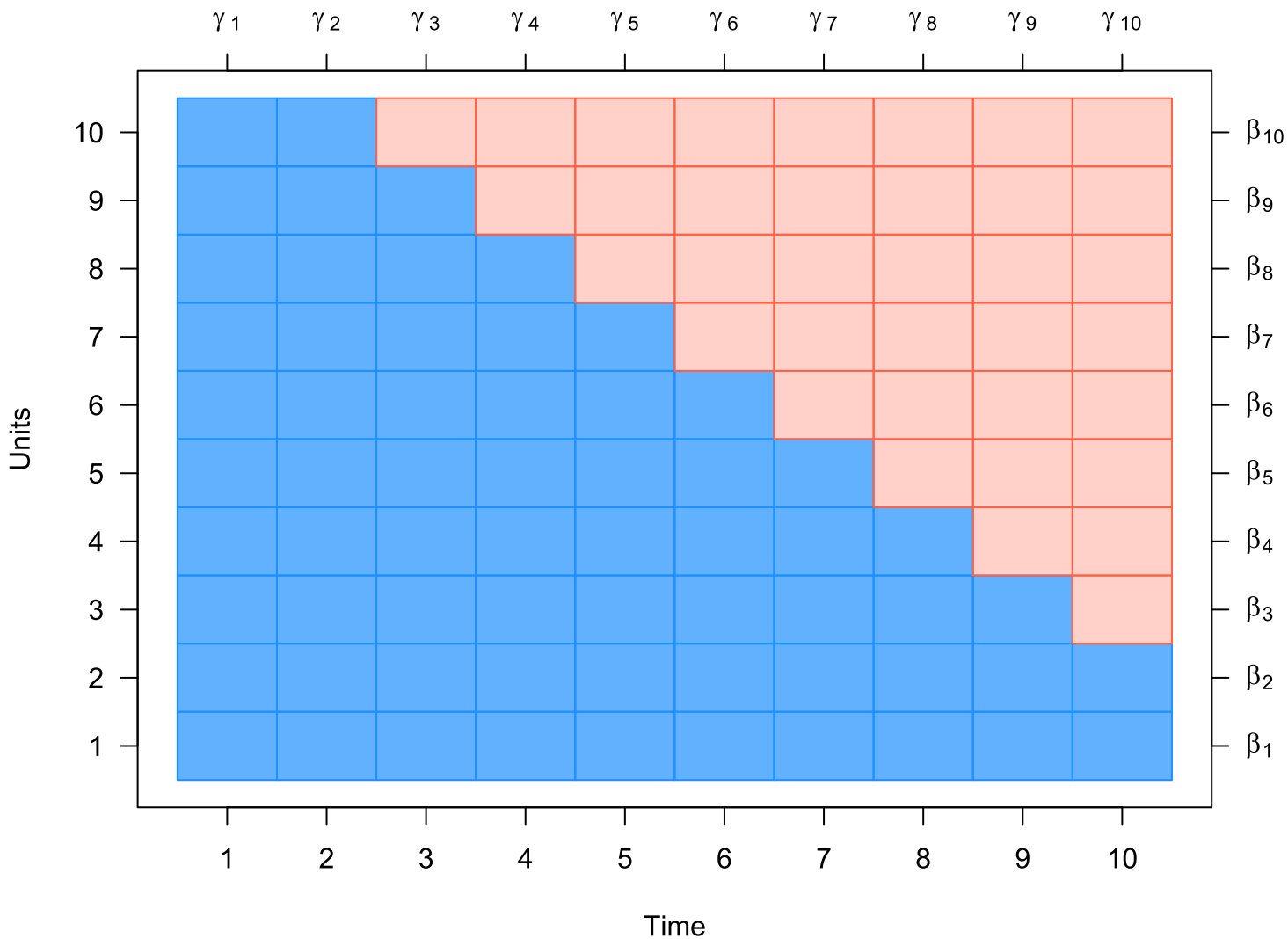
Two-Stage DiD



Two-Stage DiD



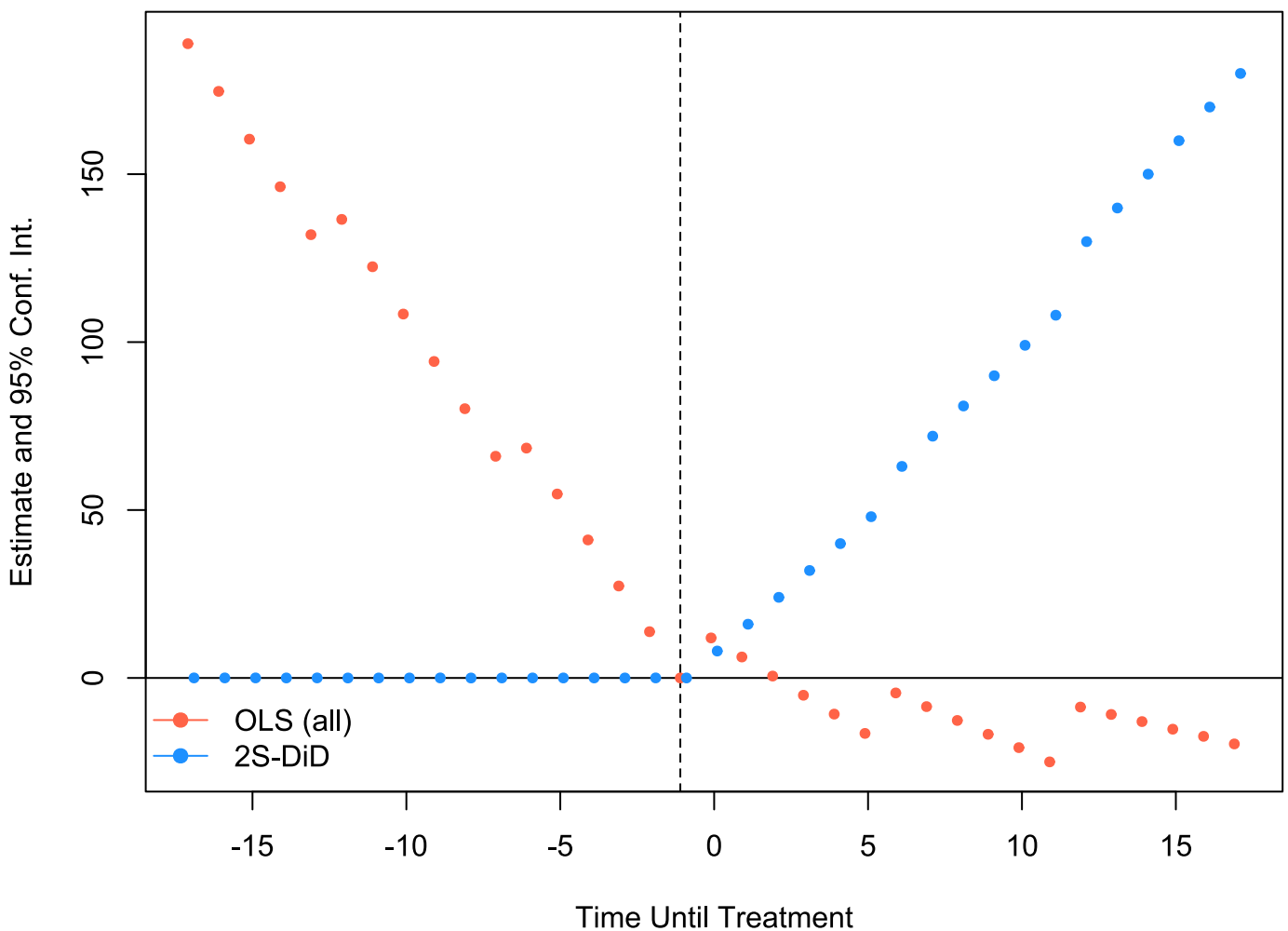
Two-Stage DiD



Two-Stage DiD

	TWFE	TWFE	2S-DiD
Treat x Post	-6.690***	26.807***	68.329***
	(0.670)	(1.110)	(1.038)
Sample	All	Y < 2004	Y < 2004
Num.Obs.	30000	24000	24000

Two-Stage DiD



Summary

Difference-in-Differences uses changes in a comparison group to estimate counterfactual changes in a treated group. If you believe this to be implausible, you are suggesting a violation of the parallel trends assumption.

Dynamic treatment effects can be estimated via an *event study.* Note that there is nothing special about breaking up a treatment effect by time – this can also be done by unit (or groups of units).

While TWFE allows for a flexible framework to handle staggered treatment timing, the resulting estimates must be diagnosed for bias and potentially corrected.

fin